A Hybrid IMRT Technique for Treatment of Breast Cancer: A Dosimetric Study

A thesis report submitted to the Department of Medical Physics and Biomedical Engineering (MPBME), Faculty of Physical and Mathematical Sciences, Gono Bishwabidyalay in partial fulfillment of requirement for the degree of Master of Science in Medical Physics and Biomedical Engineering.

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Declaration

It is hereby declared that this thesis has been conducted ethically. The data and the results presented are the genuine and obtained by me during the working period. This thesis work or any part of it has not been submitted elsewhere for the award of any degree or diploma.

_____

Signature of the candidate

DEDICATED TO

My Grandfather

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LIST OF SYMBOLS

Cm	Centimeter
Mm	Millimeter
Cm^3	Cubic Centimeters
Cm^2	Square Centimeters
mGy	Milli Gray
MV	Megavolt
MU	Monitoring Unit
D	Dose
cGy	Centri Gray
S	Second

LIST OF ABBREVIATIONS

QUANTEC	Quantitative Analyses of Normal Tissue Effects in the Clinic
FB	Free Breathing
PRV	Planning Risk Volume
Hybrid IMRT	Hybrid Intensity-Modulated Radiation Therapy
LINAC	Linear Accelerator
CT	Computed Tomography
IMRT	Intensity-Modulated Radiation Therapy
IMAT	Intensity-Modulated Arc Therapy
IP-IMRT	Inverse Planning Intensity-Modulated Radiation Therapy
FP-IMRT	Forward Planning Intensity-Modulated Radiation Therapy
RT	Radiation Treatment
MHD	Maximum Heart Distance
CLD	Central Lung Distance
MLC	Multi-Leaf Collimator
TPS	Treatment Planning System
DVH	Dose Volume Histogram
3D	Three Dimensions
3DCRT	Three Dimensional Conformal Radiation Therapy
BCS	Breast Conservation Surgery

EBRT	External Beam Radiation Therapy
ICRU	International Commission of Radiation Units and Measurements
PMMA	Polymethyl Methacrylate
DICOM	Digital Imaging and Communications in Medicine
FFF	Flattening Filter Free
CB	Contralateral Breast
GTV	Gross Tumor Volume
CTV	Clinical Target Volume
PTV	Planning Target Volume
OARs	Organ at Risks

ABSTRACT

Background and purpose: In Bangladesh, a huge number of breast cancer patients are being treated using a multimodality approach of tangential beam 3D-CRT and IMRT and volumetric-modulated arc therapy (Rapid-Arc) plans. It has been seen in many case that the dose of OARs is decreasing by using Hybrid IMRT than other multimodality approach. The hybrid IMRT plan requiring less MUs can achieve dosimetric equivalence to VMAT. The purpose of this study is to investigate the cardio-pulmonary doses, left lung parameters (V5, V10, V20 and V30) and contralateral breast respectively in Hybrid-IMRT plans according to RTOG, EMAMI and QUANTEC protocol for the left- sided breast radiotherapy.

Materials and Methods: More than eight patients with left breast carcinoma who underwent whole breast irradiation were planned using Hybrid-IMRT. Target and critical structures were delineated by one radiation oncologist according to protocol. Hybrid IMRT plans were created for prescription of 40.5Gy in 15 fractions. 3DCRT tangential open fields delivered 80 percent of the prescribed dose and two IMRT fields delivered 20 percent of the prescribed dose. IMRT fields were optimized for open field base doses, MLC was used in the two open fields to block OARs as much as possible without compromising PTV coverage. Three open fields and two IMRT fields with single iso-center were used in node positive cases. And two tangential fields and two IMRT fields with single iso-center techniques were used in node negative cases. Critical structures doses were recorded from the DVH.

Results: Hybrid IMRT average mean heart dose was (1.64 ± 0.55) Gy, V5(%) was 3.55 ± 2.65, V20 (%) was 2.5 ± 0.55 & V30 (%) was 1.05 ± 0.65 respectively. Hybrid IMRT average mean left lung dose was (7.5Gy ± 2.24) Gy. V5 (%) was 28% ± 4.5%, V10 was 20.24% ± 5.6%, V20 (%) was 14.77% ± 6.9% and V30(%) was 11.98% ± 2.8% respectively. However, hybrid IMRT offered excellent results comparable to that of full IMRT.

Conclusion: In context of limiting mean and low dose to lung and heart, H-IMRT is found to achieve good OAR sparing with acceptable PTV coverage for free breathing, left breast irradiation. The most important advantage of H-IMRT is treatment delivery time better than IMRT.

Keywords: Breast cancer radiotherapy; H-IMRT and Organs at Risk.

Chapter 1: Introduction

1.1 General Introduction

Breast Cancer is the most common malignant neoplasm in women and the main cause of death among women worldwide. Nearly 1.2 million women are diagnosed with breast cancer all over the world every year, and 500,000 women die of it each year. [1] Therefore, the breast cancer remains the primary cause of cancer mortality in women after lung cancer. With the development of treatment strategies, an enhanced awareness of medical examination, and the popularity of breast cancer screening, the 5-year overall survival of breast cancer patients has improved in the last 30 years [1]. The 5-year, 10-year, and 15-year overall survival is 89%, 83%, and 78%, respectively. Analyses of randomized clinical trials have proven that breast-preserving surgery achieves survival equivalent to mastectomy in patients with early-stage breast cancer [2], [3], [4]. Adjuvant whole-breast irradiation is performed on these patients as the standard of care. In Bangladesh, the rate of breast cancer occurrence is estimated to be 22.5 per 100,000 females of all ages, compared to 124.8 per 100,000 females worldwide. For Bangladeshi women aged between 15-44 years, breast cancer has the highest rate of occurrence 19.3 per 100,000, compared to any other type of cancer;(2013/01/20/breast-cancer-in-Bangladesh). Breast cancer can be treated using a multimodality approach of surgery, chemotherapy, radiotherapy and targeted therapy. The treatment options vary as per the stage and grading of the tumor. The most adopted treatment method for breast cancer patients is breast conservation surgery (BCS), or mastectomy followed by adjuvant radiotherapy. Adjuvant radiotherapy improves local control and improves overall survival.[5] The adjuvant radiotherapy of the left chest wall is commonly delivered by three dimensional conformal radiotherapy (3DCRT) with field-in-field technique which have shown that increased cardiac morbidity and mortality in patients treated with radiotherapy for left-sided breast cancer compared to right-sided, due to the higher cardiac dose for the left-sided patients.[6] Cardiotoxicity is most frequently reported as decreased myocardial function or coronary artery disease (also reported as ischemic heart disease or decreased cardiac perfusion). However, less common toxicities can include myocardial infarction, congestive heart failure, pericarditis, arrhythmias, angina, or valve dysfunction [7]. However, many studies have demonstrated increased cardiac mortality and morbidity after breast radiotherapy [8]. Although many of the studies included in this review involved old treatment techniques, which probably delivered a higher dose to the heart than seen in modern radiotherapy clinics, the issue of cardiac morbidity and mortality after breast cancer treatment is still relevant [9]. Dose–response curves for cardiac mortality derived from two randomized trials of surgery with or without radiotherapy for early breast cancer show a strong correlation with the dose delivered and a lesser dependency on the volume of the heart irradiated, assuming homogeneous radio-sensitivity within the irradiated heart volume [10]. The avoidance of hot spots throughout the breast volume is difficult with external whole breast radiotherapy using conventional forward wedge planning [11,12]. For this reason, intensity-modulated radiotherapy (IMRT) is gradually replacing wedge planning [13]. The development IMRT provides excellent dose homogeneity throughout the breast volume [14] that is impossible for 3D conformal radiotherapy. IMRT reduces the volume of OAR such as the heart and lung receiving high dose of radiation while improving the conformity and homogeneity of the tumor doses. The main disadvantage of IMRT is that the IMRT plans might be more susceptible to setup and motion uncertainties in breast cancer

[15-19]. The possible solution to reduce the effects of geometrical uncertainties would be the use of a hybrid technique that incorporates a combination of glancing open fields and inverse-planned IMRT beams. The development of novel hybrid techniques that reduce normal tissue injury without compromising the radiation doses to the tumor in a short delivery time is now the subject of much interest in radiation therapy research. The Hybrid IMRT has the potential of lowering the radiation doses to the OAR while improving the conformity and homogeneity to the tumor with fewer monitor units (MUs) and a short delivery time compared with IMRT. Consequently, this study can be proposed to a new hybrid treatment technique that encompasses non-modulated tangential fields as well as IMRT. A dosimetric evaluation was done with regard to target coverage as well as organ-at-risks (OAR) doses. This study was performed in dosimetric evaluations to assess the extent of dose reduction to the heart, contralateral lung and ipsilateral lung.

1.2 Aims of the Study

1.2.1 General Aim

- To produce a treatment plan that is highly conformal, a significant improvement in homogeneity, and robustness of the dose distributions against geometrical uncertainties associated with full IMRT or 3DCRT in breast radiotherapy techniques.

1.2.2. The Specific Aims

- To investigate dose to the heart, lung and contralateral breast according to protocols.
- To significantly reduce hot spot regions within the breast tissue.
- To produce less skin reaction.
- To minimize the cost of treatment of breast cancer using Hybrid-IMRT than IMRT.
- To suggest using Hybrid-IMRT technique in clinical purpose

1.3 Scope of study

Hybrid -IMRT program and the practical work were done in the Department of Radiation Oncology Nepal Cancer Hospital and Research Center, Harisiddhi Lalitpur and Department of Oncology & Radiotherapy Centre, Square Hospitals Limited and Gono Bishwabidyalay. The required features and data are collected from Varian Eclipse Treatment Planning System **(V13.7).**

1.4 Literature reviews

This literature review consists of research about left-sided breast cancer patients that receive radiation therapy with the 3DCRT, IMRT, Hybrid IMRT techniques. Various methods of radiotherapy that incorporate Hybrid-IMRT were reviewed along with when it is appropriate to use Hybrid-IMRT in a patient's treatment. Breast cancer patients may develop cardiac disease due to radiation exposure even if cured of the cancer. Although there is a relationship between breast cancer radiotherapy followed by cardiac disease the connection between specific cardiac volumes and heart disease has not been clearly documented.

Many retrospective studies have been documented that show a connection between radiation therapy to the breast and heart disease.

In the article, Rahbin et al [20] has described that treatment plans of 20 patients with left-sided breast cancer, 10 post-mastectomy treated to a prescribed dose of 45 Gy to the chest wall in 20 fractions, and 10 post-breast-conserving surgeries to a prescribed dose of 50 Gy to the whole breast in 25 fractions, with 3D-CRT were selected. The FiF-FP-IMRT plans were created by combining two open fields with three to four segments in two tangential beam directions. Eight different beam directions were chosen to create IP-IMRT plans and were inversely optimized. The homogeneity of dose to planning target volume (PTV) and the dose delivered to heart and contralateral breast were compared among the techniques in all the 20 patients. All the three radiotherapy techniques achieved comparable radiation dose delivery to PTV-95% of the prescribed dose covering > 95% of the breast PTV. The mean volume of PTV receiving 105% (V 105) of the prescribed dose was 1.7% (range 0-6.8%) for IP-IMRT, 1.9% for FP-IMRT, and 3.7% for 3D-CRT. The homogeneity and conformity indices (HI and CI) were similar for 3D-CRT and FP-IMRT, whereas the IP-IMRT plans had better conformity index at the cost of less homogeneity. The 3D-CRT and FiF-FP-IMRT plans achieved similar sparing of critical organs. The low-dose volumes (V 5Gy) in the heart and lungs were larger in IP-IMRT than in the other techniques. The value of the mean dose to the ipsilateral lung was higher for IP-IMRT than the values for with FiF-FP-IMRT and 3D-CRT. In the current study, the relative volume of contralateral breast receiving low doses (0.01, 0.6, 1, and 2Gy) was significantly lower for the FiF-FP-IMRT and 3D-CRT plans than for the IP-IMRT plan. Compared with 3D-CRT and IP-IMRT, FiF-FP-IMRT proved to be a simple and efficient planning technique for breast irradiation. It provided dosimetric advantages, significantly reducing the size of the hot spot and minimally improving the coverage of the target volume. In addition, it was felt that FiF-FP-IMRT required less planning time and easy field placements.

Gamal Elhusseiny, MD et al [21] demonstrated that IMRT becomes more suitable for treating complex target volumes as it is able to generate more complicated dose distributions and dose gradients with narrower margins than those allowed using traditional methods. As a consequence, IMRT can provide a clinical benefit in terms of increased tumor control through escalated dose and reduced normal tissue complications through OARs sparing. Gamal Elhusseiny, MD et al described CT studies of 20 patients with early stage left breast cancer who underwent breast conservative surgery were planned using 3DCRT and IMRT plans. PTV, lungs, heart and contralateral breast (CB) were outlined. 50 Gy was prescribed to the isocenter. Dose volume histogram parameters (DVPs) for both plans were compared and analyzed statistically using Wilcoxon Signed-Ranks test of SPSS (version 18). Both 3DCRT and IMRT achieved adequate and comparable target coverage. On the other hand, tangential IMRT showed insignificant higher PTV maximum dose with significant less dose homogeneity compared to 3DCRT. This is with greater sparing of the lungs, heart and contralateral breast in IMRT plans.

Sherif Elzawawy et al. [22] compared the dosimetry for the left breast cancer radiotherapy using three different radiotherapy techniques, tangential wedged fields (TW), segmented field (FiF) and inverse planning IMRT (IP-IMRT). They demonstrated that Twenty patients have undergone left breast-conservative surgery and received a prescribed dose of 50Gy /25 fractions. Results: The mean PTV receiving >105% (V105) dose was1.75% for IP-IMRT,

2.03% for FiF, and 4.82% for TW. The mean V95% was 92.1% for TW, 96% for FiF, and 95.1% for IP-IMRT; these differences regarding V105% and V95% are statistically significant through paired comparison between FiF vs TW and IP-IMRT vs TW, with no statistically significant difference between FiF and IP-IMRT. Better conformity and homogeneity indices for FiF and IMRT compared to TW with statistical significant difference. Regarding organs at risk, left lung and heart have higher values of V5, V10, and V20 for IP-IMRT compared to TW and FiF; the differences are statistically significant, lower coronary artery regionV30 vules for IPIMRT compared to TW and FiF but no difference in the Dmean between IPIMRT and FIF. FiF and TW decrease the contralateral breast dose significantly compared to IP IMRT.

Suresh Moorthy et al. [23] demonstrated that SIB-3DCRT treatment plans were created for 36 consecutive patients. Dose was prescribed as 45 Gy in 25 fractions to the planning target volume (PTV)-1 and 60 Gy in 25 fractions to PTV-2. Treatment plans were normalized to 95% of PTV volume receiving 95% of the prescription dose. The conformity index (CI), homogeneity index (HI), lung dose, heart dose, left anterior descending artery(LAD) dose, and low dose volume and integral dose of normal healthy tissue were recorded and analyzed. Results: With the use of IMRT technique, there was an improvement in CI (0.14) when compared to CI of 3DCRT (0.18; $p = 0.01$). However, there was no significant difference in the HI ($p = 0.45$). On average, the V20Gy of ipsilateral lung was 37.9 % for 3DCRT and 22.4 % ($p < 0.01$) for IMRT, whereas the V20Gy of total lung (ipsilateral + contralateral) was 21.8% for 3DCRT and 12.14 ($p < 0.01$) for IMRT. Similarly, average V40Gy of heart was 7.5 % for 3DCRT and 2.13 % ($p = 0.01$) for IMRT. The LAD maximum dose to left side breast patients, on average, was 39.5 Gy for 3DCRT and 29.17Gy ($p = 0.03$) for IMRT. The average number of monitor units was about 180 for 3DCRT and 1441 ($p < 0.01$) for IMRT. IMRT for breast cancer treatment is feasible.In comparison to 3DCRT. IMRT can reduce the maximum dose to the target volume, and dose to the OAR. However, 3DCRT technique is superior in terms of low dose volume, integral dose, and treatment time. With the use of breath-hold gated technique in IMRT, it can further improve the target coverage and reduction of doses to the heart, lung, and LAD. SIB technique could reduce the overall treatment duration by about one week.

Azza Helal et al. [24] have demonstrated that various radiation techniques include Three Dimensional Conformal Radiotherapy (3DCRT), breath hold and Intensity Modulated Radiotherapy (IMRT) are used to irradiate the breast and to limit the dose to the heart and lungs. IMRT is used as a class solution for cardiac protection in left sided breast cancer patients. They described that CT simulation was done for thirty left sided breast cancer patients who underwent conservative surgery. Planning target volume (PTV), heart and ipsilateral lung were contoured. 3DCRT and IMRT plans were generated for 50Gy using opposed tangential medial and lateral photon beams. Five segments were generated for each IMRT field. Heart V5Gy, V20Gy, V30Gy and V40Gy and ipsilateral lung V20Gy, mean dose, PTV D95%, PTV Dmax and dose homogeneity index were compared using Wilcoxon Signed-Ranks test. IMRT plans achieved significant reduction of 12%, 15%, 21% & 53% in heart V5Gy, V20Gy, V30Gy and V40Gy and of 18% & 11% in ipsilateral lung mean dose and V20Gy compared with 3DCRT (p values <0.001). PTV coverage was adequate and comparable in both plans (P for PTV D95% & Dmax% =0.188 & 0.553). The dose homogeneity within PTV was significantly better with 3DCRT (P=0.004). Tangential beam

IMRT in left sided breast cancer leads to a significant dose reduction in heart and ipsilateral lung with comparable target coverage compared to tangential beam 3DCRT.

Xiaoxue Xie et al. [25] compared the dosimetric characteristics of left-sided whole breast irradiation among 3-dimensional conformal radiotherapy (3D-CRT), 4-field inverse-planned intensity-modulated radiotherapy (IP-IMRT) and hybrid IMRT technique (combining 3D-CRT beams and IP-IMRT beams) with respect to target coverage and irradiation of organs at risk. The 3 different planning techniques were analyzed for 8 patients with left-sided breast conserving surgery. Plans were compared on the basis of planning target volume (PTV) dose conformity, homogeneity and the volumes of normal tissues treated based on dose-volume histograms (DVHs). DVHs were calculated for the PTV, heart, and the bilateral lungs, contralateral breast, and soft tissue surrounding the breast PTV (VOB) volume. IP-IMRT and hybrid IMRT techniques comparably improved the PTV dose homogeneity and conformity (CI) significantly, compared to the conventional 3D-CRT technique ($P<0.017$); the IP-IMRT technique only could additionally benefit patients by decreasing the high-dose (40 Gy) volume for heart and ipsilateral lung compared with the hybrid IMRT technique ($P<0.017$); the hybrid IMRT plans achieved a further improvement by compromising the increase of low-dose volume (total lung V13, contralateral lung V5, heart V10 and soft tissue surrounding the breast V5) compared with IP-IMRT plans ($P<0.017$). Hybrid IMRT plans achieved equivalent PTV dose uniformity to IP-IMRT plans and compromised the low-dose volume and requirement of clinic resource between IP-IMRT and 3D-CRT plans, promoting it as a standard practice of left-sided breast irradiation for patients in good-ordered cardiopulmonary health.

Volker Rudat1 et al. [26] have demonstrated that this study evaluates the dose distribution of reversed planned tangential beam intensity modulated radiotherapy (IMRT) compared to standard wedged tangential beam three-dimensionally planned conformal radiotherapy (3D-CRT) of the chest wall in unselected postmastectomy breast cancer patients. They described that 20 unselected subsequent post-mastectomy breast cancer patient's tangential beam IMRT and tangential beam 3D-CRT plans were generated for the radiotherapy of the chest wall. The prescribed dose was 50 Gy in 25 fractions. Dose-volume histograms were evaluated for the PTV and organs at risk. Parameters of the dose distribution were compared using the Wilcoxon matched pairs test. Tangential beam IMRT statistically significantly reduced the ipsilateral mean lung dose by an average of 21% (1129 cGy versus 1437 cGy). In all patients treated on the left side, the heart volume encompassed by the 70% isodose line (V70%; 35 Gy) was reduced by an average of 43% (5.7% versus 10.6%), and the mean heart dose by an average of 20% (704 cGy versus 877 cGy). The PTV showed a significantly better conformity index with IMRT; the homogeneity index was not significantly different. Tangential beam IMRT significantly reduced the dose-volume of the ipsilateral lung and heart in unselected postmastectomy breast cancer patients.

Mudasir Ashraf et al. [27] have described a comparison between the dosimetric aspects of 3DCRT and IMRT in the whole breast radiotherapy. Both right and left sided computed tomography simulations of 14 women with early stage breast cancer were used for our retrospective study to compare the 3DCRT and IMRT. The dose prescribed was 50 Gray in 25 fractions to the whole breast PTV. The PTV was defined by adding unequal margins to the directional safety margin status of each lumpectomy cavity (i.e., medial, lateral, superior, inferior and deep margins measured from the tumor front after the examination of the

surgical specimen: 2, 1.5, and 1 cm for resection margins < 1 cm, 1-2 cm, and > 2cm, respectively) and then modified so that it was no longer closer than 3 mm to the skin surface and was no deep than the lung –chest interface. The prescribed dose delivered in 5 fractions per week schedule. Treatment plans were compared for target minimum dose, maximum dose, mean dose, conformity index, heterogeneity index and doses to organs at risk were compared and analyzed. Results: The target coverage was achieved with 90% prescription to the 95% volume of the PTV. Conformity to the PTV was significantly higher with 3DCRT technique than IMRT. 3DCRT technique seems better in sparing critical organs parameters like lung V20 and Mean, heart, V25, Maximum, both lungs V20, Mean and Dose to the Normal Healthy tissue3DCRT produces better conformity and heterogeneity indices of the target volume, also reduces dose to OARs and reduces the risk of radiation induced heart diseases.

Charles S Mayo et al. [28] have evaluated a hybrid intensity modulated radiation therapy (IMRT) technique as a class solution for treatment of the intact breast. The following five plan techniques were compared for 10 breast patients using dose–volume histogram analysis: conventional wedged-field tangents (Tangents), forward-planned field-within- field tangents (FIF), IMRT-only tangents (IMRT tangents), conventional open plus IMRT tangents (4-field hybrid), and conventional open plus IMRT tangents with 2 anterior oblique IMRT beams (6-field hybrid). The 4-field hybrid and FIF achieved dose distributions better than Tangents and IMRT tangents. The volume of tissue outside the planning target volume receiving >110% of prescribed dose was largest for IMRT tangents (average 158 cc) and least for 6-field hybrid (average 1 cc); the FIF and 4-field hybrid were comparable (average 15 cc). Heart volume >30 Gy averaged 13 cc for all techniques, except Tangents, for which it was 32 cc. Average total lung volume >20 Gy was 7% for all. Contralateral breast doses were <3% for all. Planning time for hybrid techniques was significantly less than for conventional FIF technique. The 4-field hybrid technique is a viable class solution. The 6-field hybrid technique creates the most conformal dose distribution at the expense of more normal tissue receiving low dose.

In a study ref. [29] have compared the dosimetric characteristics of left-sided whole breast irradiation among 3-dimensional conformal radiotherapy (3D-CRT), 4-field inverse-planned intensity-modulated radiotherapy (IP-IMRT) and hybrid IMRT technique (combining 3D-CRT beams and IP-IMRT beams) with respect to target coverage and irradiation of organs at risk. The 3 different planning techniques were analyzed for 8 patients with left-sided breast conserving surgery. Plans were compared on the basis of planning target volume (PTV) dose conformity, homogeneity and the volumes of normal tissues treated based on dose-volume histograms (DVHs). DVHs were calculated for the PTV, heart, and the bilateral lungs, contralateral breast, and soft tissue surrounding the breast PTV (VOB) volume. IP-IMRT and hybrid IMRT techniques comparably improved the PTV dose homogeneity and conformity (CI) significantly, compared to the conventional 3D-CRT technique ($P<0.017$); the IP-IMRT technique only could additionally benefit patients by decreasing the high-dose (40 Gy) volume for heart and ipsilateral lung compared with the hybrid IMRT technique ($P<0.017$); the hybrid IMRT plans achieved a further improvement by compromising the increase of low-dose volume (total lung V13, contralateral lung V5, heart V10 and soft tissue surrounding the breast V5) compared with IP-IMRT plans ($P<0.017$). Hybrid IMRT plans achieved equivalent PTV dose uniformity to IP-IMRT plans and compromised the low-dose

volume and requirement of clinic resource between IP-IMRT and 3D-CRT plans, promoting it as a standard practice of left-sided breast irradiation for patients in good-ordered cardiopulmonary health.

Farace P et al. [30] have described between tangential and not-tangential hybrid intensity modulated radiation therapy (IMRT) for whole-breast irradiation Seventy-eight (36 right-, 42 left-) breast patients were randomly selected. Hybrid IMRT was performed by direct aperture optimization. A semi-automated method for planning hybrid IMRT was implemented using Pinnacle scripts. A plan optimization volume (POV), defined as the portion of the planning target volume covered by the open beams, was used as the target objective during inverse planning. Treatment goals were to prescribe a minimum dose of 47.5 Gy to greater than 90% of the POV and to minimize the POV and/or normal tissue receiving a dose greater than 107%. When treatment goals were not achieved by using a 4-field technique (2 conventional open plus 2 IMRT tangents), a 6-field technique was applied, adding 2 non tangential (anterior-oblique) IMRT beams. Using scripts, manual procedures were minimized (choice of optimal beam angle, setting monitor units for open tangentials, and POV definition). Treatment goals were achieved by using the 4-field technique in 61 of 78 (78%) patients. The 6-field technique was applied in the remaining 17 of 78 (22%) patients, allowing for significantly better achievement of goals, at the expense of an increase of low-dose (~5 Gy) distribution in the contralateral tissue, heart, and lungs but with no significant increase of higher doses (~20 Gy) in heart and lungs. The mean monitor unit contribution to IMRT beams was significantly greater (18.7% vs 9.9%) in the group of patients who required 6-field procedure. Because hybrid IMRT can be performed semi-automatically, it can be planned for a large number of patients with little impact on human or departmental resources, promoting it as the standard practice for whole-breast irradiation.

In similar study ref. [31] demonstrated a conformal planning for post-mastectomy radiotherapy (PMRT) with nodal regions with an intention to overcome the treatment planning limitations imposed by conventional techniques. Twenty-four patients who underwent surgical mastectomy followed by PMRT were included in this study. Initially, a treatment plan comprising tangential beams fitted to beam's-eye-view (BEV) of chest wall (CW) and a direct anterior field fitted to BEV of nodal region, both sharing a single isocenter was generated using Eclipse treatment planning system. Multiple field-in-fields with optimum beam weights (5% per field) were added primarily from the medial tangent, fitted to BEV of entire target volume, and finally converted into a dynamic portal. Dosimetric analysis for the treatment plans and fluence verification for the dynamic portals were performed. Conformal plans with SDP showed excellent dose coverage (V >95%), higher degree of tumor dose conformity (≤1.25) and homogeneity (≤0.12) without compromising the organ at risk sparing for PMRT with nodal region. Treatment plans with SDP considerably reduced the lower isodose spread to the ipsilateral lung, heart, and healthy tissue without affecting the dose homogeneity. Further, gamma evaluation showed more than 96% pixel pass rate for standard 3%/3 mm dose difference and distance-to-agreement criteria. Moreover, this plan offers less probability of “geometrical miss” at the highly irregular CW with regional nodal radiotherapy. Hybrid conformal plans with SDP would facilitate improved dose distribution and reduced uncertainty in delivery and promises to be a suitable treatment option for complex post-mastectomy CW with regional nodal irradiation.

Ahmad Amoush et. al. [32] have compared the single-isocenter, four field hybrid IMRT with the two-isocenter techniques to treat the whole breast and supraclavicular fields and to investigate the intra-fraction motions in both techniques in the superior direction. Fifteen breast cancer patients who underwent lumpectomy and adjuvant radiation to the whole breast and supraclavicular (SCV) fossa at our institution were selected for this study. Two planning techniques were compared for the treatment of the breast and SCV lymph nodes. The patients were divided into three subgroups according to the whole breast volume. For the two-isocenter technique, conventional wedged or field-within-a-field tangents (FIF) were used to match with the same anterior field for the SCV region. For the single-isocenter technique, four-field hybrid IMRT was used for the tangent fields matched with a half blocked anterior field for the SCV region. To simulate the intra-fraction uncertainties in the longitudinal direction for both techniques, the treatment isocenters were shifted by 1 mm and 2 mm in the superior direction. The average breast clinical tumor volume (CTV) receiving 100% (V100%) of the prescription dose (50 Gy) was 99.3% ± 0.5% and 96.4% ± 1.2% for the for two-isocenter and single-isocenter plans ($p < 0.05$), respectively. The breast CTV receiving 95% of the prescription dose (V95%) was close to 100% in both techniques. The average breast CTV receiving 105% (V105%) of the prescription dose was 32.4% ± 19.3% and 23.8% ± 13.3% ($p = 0.08$). The percentage volume of the breast CTV receiving 110% of the dose was 0.4% ± 1.2% in the two-isocentric technique vs. 0.1% ± 0.2% in the single-isocentric technique. The average uniformity index was 0.91 ± 0.02 vs. 0.91 ± 0.01 in both techniques ($p = 0.04$), but had no clinical impact. The percentage volume of the contralateral breast receiving a dose of 1 Gy was less than 2.3% in small breast patients and insignificant for medium and large breast sizes. The percentage of the total lung volume receiving > 20 Gy (V20Gy) and the heart receiving > 30 Gy (V30Gy) were 13.6% vs. 14.3% ($p = 0.03$) and 1.25% vs. 1.2% ($p = 0.62$), respectively. Shifting the treatment isocenter by 1 mm and 2 mm superiorly showed that the average maximum dose to 1 cc of the breast volume was 55.5 ± 1.8 Gy and 58.6 ± 4.3 Gy in the two-isocentric technique vs. 56.4 ± 2.1 Gy and 59.1 ± 5.1 Gy in the single-isocentric technique (p = 0.46, 0.87), respectively. The single-isocenter technique using four-field hybrid IMRT approach resulted in comparable plan quality as the two-isocentric technique. The single-isocenter technique is more sensitive to intra-fraction motion in the superior direction compared to the two-isocentric technique. The advantages of the single-isocenter include elimination of isocentric errors due to couch and collimator rotations and reduction in treatment time. This study supports consideration of a single-isocenter four-field hybrid IMRT technique for patients undergoing breast and supraclavicular nodal irradiation.

OUYANG Shuyu et. Al. [33] evaluated the potential dosimetric benefits and optimal indications of intensity modulated radiation therapy (IMRT) and hybrid intensity modulated radiation therapy (Hybrid IMRT) for the left side breast cancer patients after breast-conservation therapy. Eight patients with left breast carcinoma who received breast-conservation surgery were selected for this study. Two plans were designed in 3-dimensional treatment planning system. The dose distributions of target volume and normal tissues, conformal index (CI) and heterogeneous index (HI) were analyzed by dose volume histogram (DVH). The PTV coverage was the same in the two radiotherapy plans. A better dose uniformity throughout the whole breast in Hybrid IMRT plan was achieved. The CI, the percentage of PTV receiving more than 105% prescribed dose (V105%), the percentage of PTV receiving more than 110% prescribed dose (V110%), and the Dmax, Dmin and Dmean

of PTV were similar in the two plans. The study was compared with the Hybrid IMRT and IMRT: V13 of the ipsilateral lung decreased from 27.66% to 20.7%, V5 of the contralateral lung decreased from 8.01% to 2.25%, V10 and V20 of the heart decreased from 35.23% and 16.77% to 19.22% and 10.6% respectively, V5 and V10 of the contralateral breast decreased from 35% and 10.39% to 20.38% and 5.7% respectively, all with significant difference. V30 and V40 of the ipsilateral lung and V40 of the heart increased by 1.28%, 1.48%, and 2.48%, with significant difference. Hybrid IMRT is a better choice for patients whose treatment position is inaccurate or cannot be repeated well.

Yuan-Gui Chen et. al. [34] have demonstrated the feasibility and advantages of a hybrid, volumetric arc therapy technique that used two 90° coplanar arcs and two three-dimensional conformal tangential beams in the simultaneous-integrated boost radiotherapy of left-sided breast cancer after breast-conserving surgery. A total of nine patients with stage I, left-sided breast cancer who underwent breast-conserving surgery were selected for this retrospective study. For each patient, a hybrid arc plan was generated and then compared with two hybrid intensity modulated radiotherapy plans. All plans were optimized using the same objectives and dose constraints. The prescription dose was 50.4 Gy to the planning target volume with simultaneous boost to 60 Gy to the expanded gross target volume in 28 fractions. The differences among these hybrid plans were analyzed by the Kolmogorov–Smirnov test or the Wilcoxon rank sum test. The hybrid arc plans achieved the clinical requirements of target dose coverage and normal tissue (NT) dose constraints. It was found that the hybrid arc plans showed advantages in the conformity index of the expanded gross target volume, the V5of the heart, the D2of the left ventricle, and the D2and V50.4of NTs. The average beam-on time and monitor units of the hybrid arc plans were significantly lower ($P < 0.001$).

Osamu Takahashi**1** et. al. [35] have demonstrated to quantify the target coverage, homogeneity, and robustness of the dose distributions against geometrical uncertainties associated with four whole breast radiotherapy techniques. The study was based on the planning-computed tomography-datasets of 20 patients who underwent whole breast radiotherapy. A total of four treatment plans (wedge, field-in-field [FIF], hybrid intensity-modulated radiotherapy [IMRT], and full IMRT) were created for each patient. The hybrid IMRT plans comprised two opposed tangential open beams plus two IMRT beams. Setup errors were simulated by moving the beam isocenters by 5 mm in the anterior or posterior direction. With the original plan, the wedge technique yielded a high volume receiving ≥107% of the prescription dose (V107; 7.5%±4.2%), whereas the other three techniques yielded excellent target coverage and homogeneity. A 5 mm anterior displacement caused a large and significant increase in the V107 (+5.2%±4.1%, $p<0.01$) with the FIF plan, but not with the hybrid IMRT (+0.4%±1.2%, $p=0.11$) or full IMRT (+0.7%±1.8%, $p=0.10$) plan. A 5-mm posterior displacement caused a large decrease in the V95 with the hybrid IMRT (-2.5%± 3.7%, $p<0.01$) and full IMRT (-4.3%±5.1%, $p<0.01$) plans, but not with the FIF plan (+0.1%±0.7%, $p=0.74$). The decrease in V95 was significantly smaller with the hybrid IMRT plan than with the full IMRT plan ($p<0.01$). The FIF, hybrid IMRT, and full IMRT plans offered excellent target coverage and homogeneity. Hybrid IMRT provided better robustness against geometrical uncertainties than full IMRT, whereas FIF provided comparable robustness to that of hybrid IMRT.

D. Bertsimas et. al. [36] have described Intensity-Modulated Radiation Therapy is the technique of delivering radiation to cancer patients by using non-uniform radiation fields from

selected angles, with the aim of reducing the intensity of the beams that go through critical structures while reaching the dose prescription in the target volume. Two decisions are of fundamental importance: to select the beam angles and to compute the intensity of the beams used to deliver the radiation to the patient. Often, these two decisions are made separately: first, the treatment planners, on the basis of experience and intuition, decide the orientation of the beams and then the intensities of the beams are optimized by using an automated software tool. Automatic beam angle selection (also known as Beam Angle Optimization) is an important problem and is today often based on human experience. In this context, we face the problem of optimizing both the decisions, developing an algorithm which automatically selects the beam angles and computes the beam intensities. We propose a hybrid heuristic method, which combines a simulated annealing procedure with the knowledge of the gradient. Gradient information is used to quickly find a local minimum, while simulated annealing allows to search for global minima. As an integral part of this procedure, the beam intensities are optimized by solving a Linear Programming model. The proposed method presents a main difference from previous works: it does not require to have on input a set of candidate beam angles. Indeed, it dynamically explores angles and the only discretization that is necessary is due to the maximum accuracy that can be achieved by the linear accelerator machine. Experimental results are performed on phantom and real-life case studies, showing the advantages that come from our approach.

In the article, Shayne K. Smith, et. al. [37] have compared conventional simultaneous integrated boost (SIB) planning technique with a hybrid SIB intensity-modulated radiation therapy (IMRT) technique with varying open tangent to IMRT field dose ratios. In total, 15 patients with conventional SIB treatment plans were re-planned with hybrid SIB IMRT technique using three differing open field: IMRT dose ratios, that is, 80:20, 70:30 and 60:40. Plans were compared using dosimetric comparison of organs at risk (OARs) and homogeneity and conformity indexes across target structures. All hybrid plans reduced dose maximums and showed a reduction of high doses to both lungs but increased lower doses, that is, V5, with similar results discovered for the heart. Contralateral breast dose was shown to decrease V5 and V1 measures by hybrid arms, whereas increasing the V2. Left anterior descending artery dose and non-irradiated structures were reduced by all hybrid arms. The homogeneity and conformity increased across all hybrid arms. Qualitative assessment of all plans also favoured hybrid plans. Hybrid plans produced superior dose conformity, homogeneity, reduced dose maximums and showed an improvement in most OAR parameters. The 70:30 hybrid technique exhibited greater benefits as a class solution to the sample than conventional plans due to superior dose conformity and homogeneity to target volumes.

Jia-Fu Lin et al. [38] compared the dosimetric performance of 3 different treatment techniques: hybrid volumetric-modulated arc therapy (hybrid-VMAT), pure-VMAT, and fixed-field intensity-modulated radiation therapy (F-IMRT) for whole-breast irradiation of left-sided early breast cancer. The hybrid-VMAT treatment technique and 2 other treatment techniques—pure-VMAT and F-IMRT—were compared retrospectively in 10 patients with left-sided early breast cancer. The treatment plans of these patients were re-planned using the same contours based on the original computed tomography (CT) data sets. Dosimetric parameters were calculated to evaluate plan quality. Total monitor units (MUs) and delivery time were also recorded and evaluated. The hybrid-VMAT plan generated the best results in

dose coverage of the target and the dose uniformity inside the target ($p < 0.0001$ for conformal index [CI]; $p = 0.0002$ for homogeneity index [HI] of planning target volume [PTV] (50.4 Gy) and $p < 0.0001$ for HI of PTV (62 Gy)). Volumes of ipsilateral lung irradiated to doses of 20 Gy (V (20 Gy)) and 5 Gy (V (5 Gy)) by the hybrid-VMAT plan were significantly less than those of the F-IMRT and the pure-VMAT plans. The volume of ipsilateral lung irradiated to a dose of 5 Gy was significantly less using the hybrid-VMAT plan than that using the F-IMRT or the pure-VMAT plan. The total mean MUs for the hybrid-VMAT plan were significantly less than those for the F-IMRT or the pure-VMAT plan. The mean machine delivery time was 3.23 ± 0.29 minutes for the hybrid-VMAT plans, which is longer than that for the pure-VMAT plans but shorter than that for the F-IMRT plans. The hybrid-VMAT plan is feasible for whole-breast irradiation of left-sided early breast cancer.

Literature have shown that Hybrid-IMRT decreases the high dose to heart and ipsilateral lung, contralateral lung and contralateral breast for left breast patients compared to 3DCRT and full-IMRT. The Hybrid technique creates the most conformal dose distribution at the expense of more normal tissue receiving low dose. Hybrid-IMAT is a technique which will provide the benefit of IMRT and at the same time it will reduce the treatment time. The most important advantage of Hybrid-IMAT over fixed field IMRT was substantial reduction in treatment time.

Most of the literatures have showcased four fields or more fields with 50 Gy in 25 fractions used in hybrid IMRT for breast cancer. In this study, two fields with hypo-fractionated dose (40.5Gy in 15 fractions) have been used in the hybrid-IMRT plan for breast cancer.

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Chapter 2: Theoretical Background

2.1 Anatomy of Breast

The breast is positioned within the superficial fascia on the anterior chest wall. It extends from the second rib superiorly to the inframammary fold, typically at the sixth rib. Horizontally, the breast spans from the lateral sternal edge to the midaxillary line. Breast tissue extending into the axilla is referred to as the axillary tail of Spence. The pectoral fascia, serratus anterior, external oblique muscles, and upper rectus sheath form the posterior border of the breast. The retromammary bursa consists of loose areolar tissue located between the glandular breast elements and deep fascia of the pectoralis major. This space allows for breast mobility along the chest wall. Anteriorly, the breast is attached to the skin by fibrous connective tissue septa called suspensory or Cooper ligaments. These ligaments insert perpendicularly to the superficial fascia below the dermis and attach posteriorly to the deep fascia, providing structural support without impairing mobility. Infiltration of these ligaments by malignancy leads to skin retraction and dimpled appearance of the breast. Each breast contains 15-20 glands called lobes, where milk is produced in women who are breastfeeding. These lobes are connected to the nipple by 6-8 tubes called ducts which carry milk to the nipple. The breast and armpit also contain lymph nodes and vessels carrying lymph fluid and white blood cells, which are part of the immune system. Much of the rest of the breast is fatty tissue. The figure 2.1 shows more details on each part of the breast anatomy. Most breast carcinomas are of adenocarcinomas that originate from the terminal ducts of the breast. Lymph nodes are small, bean-shaped organs that help fight infection and are found throughout the body. They produce and filter a colorless fluid called lymph, which contains white blood cells known as lymphocytes (immune cells involved in defending against infections and such diseases as cancer). Lymph vessels filter and carry lymph fluid from the breast to the lymph nodes [40]. Clusters of lymph nodes near the breast are located in the armpit (known as axillary lymph nodes), above the collarbone, in the neck, and in the chest. The figure 2.2 shows more details on each part of the lymph nodes.

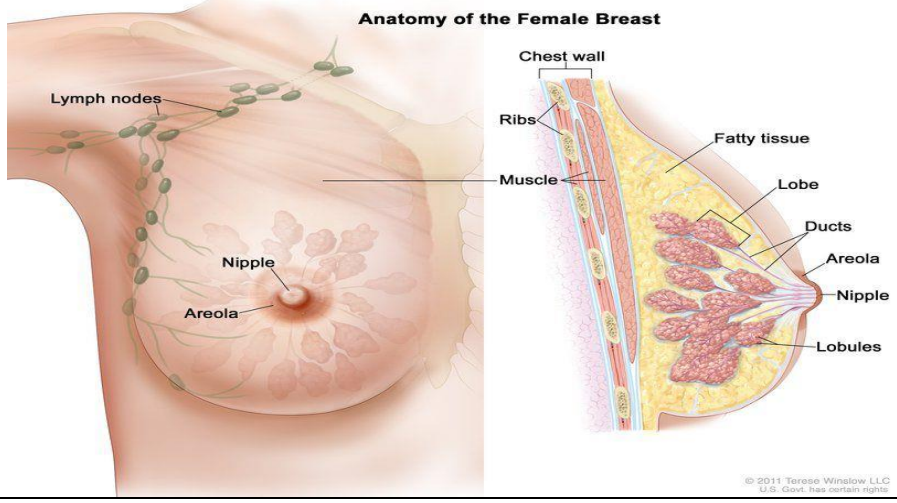


Fig. 2.1 : Diagram explains the anatomy of breast

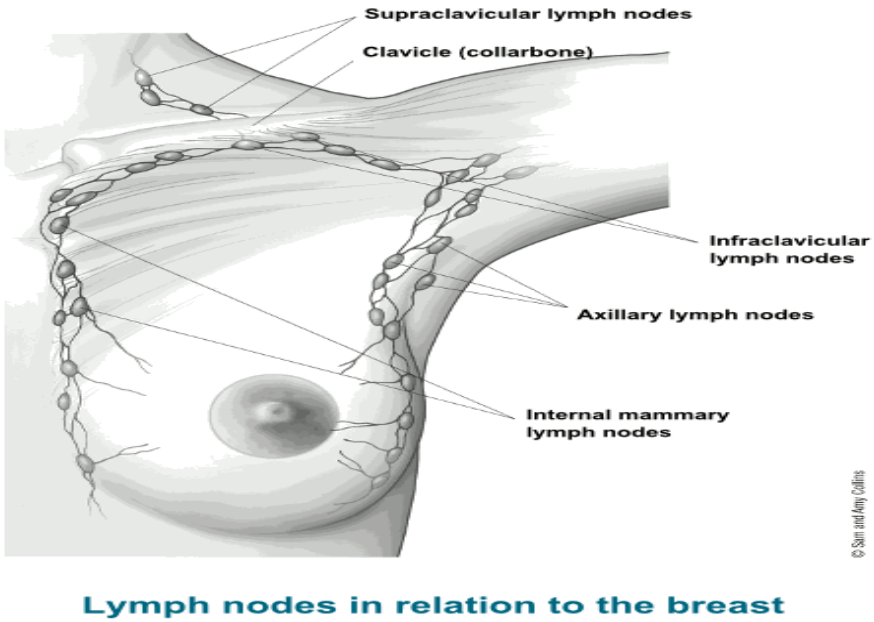


Fig. 2.2 : Diagrams explains the lymph nodes of breast

2.2 Breast Cancer

Breast cancer is an uncontrolled growth of breast cells. The term “breast cancer” refers to a malignant tumor that has developed from cells in the breast. Usually breast cancer either begins in the cells of the lobules, which are the milk-producing glands, or the ducts, the passages that drain milk from the lobules to the nipple. Less commonly, breast cancer can begin in the stromal tissues, which include the fatty and fibrous connective tissues of the breast. Breast cancer occurs almost entirely in women, but men can get breast cancer, too.

There are different signs and symptoms of breast cancer, including:

- A lump in the breast or armpit
- A change in the size or shape of the breast
- Breast pain or sore nipple
- Dimpling of the skin or thickening in the breast tissue
- A nipple that’s turned in (inverted)
- A rash on, or discharge from the nipple
- Swelling or a lump in the armpit

Figure 1.0 shows more details of the signs and symptoms of breast cancer.

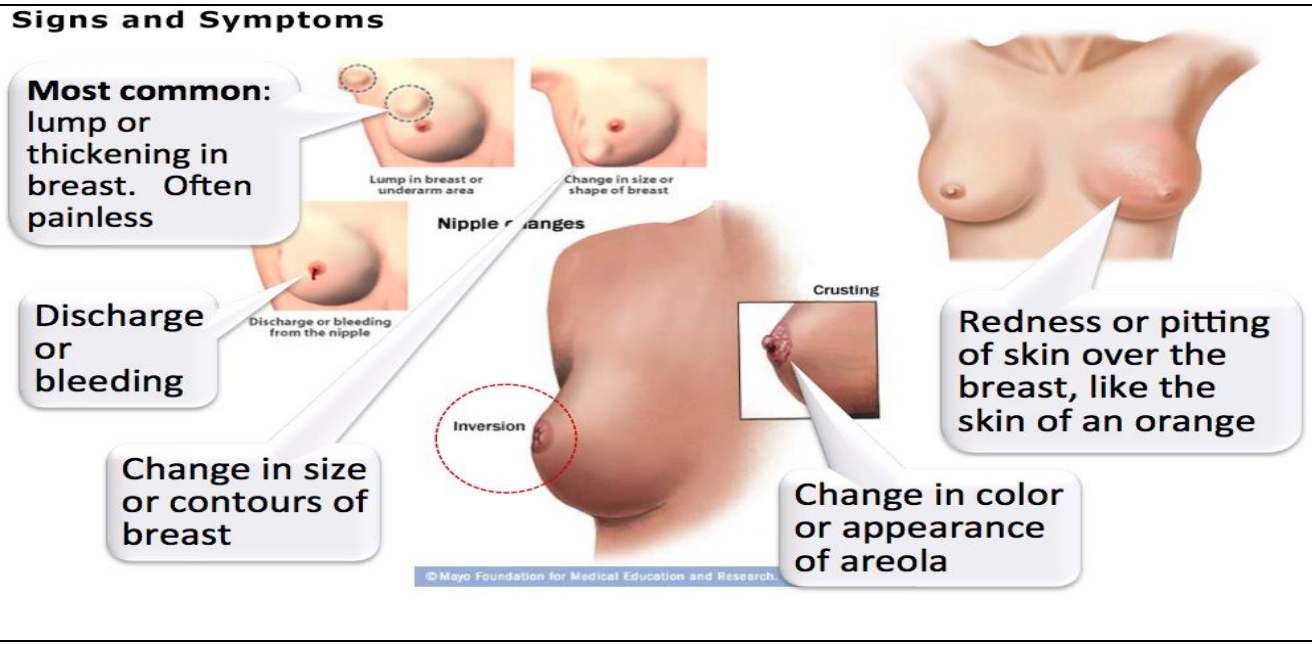


Figure 2.3 explains the signs and symptoms of breast cancer

2.2.1 Types of Breast Cancer

There are many types of breast cancer. Some are more common than others, and there are also combinations of cancers. Some of the most common types of cancer are as follows:

- **Ductal carcinoma in situ:** The most common type of noninvasive breast cancer is ductal carcinoma in situ (DCIS). This type of cancer has not spread and therefore usually has a very high cure rate.
- **Invasive ductal carcinoma:** This cancer starts in a duct of the breast and grows into the surrounding tissue. It is the most common form of breast cancer. About 80% of invasive breast cancers are invasive ductal carcinoma.
- **Invasive lobular carcinoma:** This breast cancer starts in the glands of the breast that produce milk. Approximately 10% of invasive breast cancers are invasive lobular carcinoma.

The remainder of breast cancers are much less common and include the following:

- **Mucinous carcinoma** is formed from mucus-producing cancer cells. Mixed tumors contain a variety of cell types.
- **Medullary carcinoma** is an infiltrating breast cancer that presents with well-defined boundaries between the cancerous and noncancerous tissue.
- **Inflammatory breast cancer:** This cancer makes the skin of the breast appear red and feel warm (giving it the appearance of an infection). These changes are due to the blockage of lymph vessels by cancer cells.
- **Triple-negative breast cancers:** This is a subtype of invasive cancer with cells that lack estrogen and progesterone receptors and have no excess of a specific protein

(HER2) on their surface. It tends to appear more often in younger women and African-American women.

- **Paget's disease of the nipple:** This cancer starts in the ducts of the breast and spreads to the nipple and the area surrounding the nipple. It usually presents with crusting and redness around the nipple.
- **Adenoid cystic carcinoma:** These cancers have both glandular and cystic features. They tend not to spread aggressively and have a good prognosis.
- **Lobular carcinoma in situ**: This is not a cancer but an area of abnormal cell growth that can lead to invasive breast cancer later in life.

2.2.2 Breast Cancer Screening

Screening examinations are tests performed to find disease before symptoms begin. The goal of screening is to detect disease at its earliest and most treatable stage. In breast cancer screening, a woman who has no signs or symptoms of breast cancer undergoes a breast examination such as:

- **Clinical breast exam:** A physical examination of the breast by a doctor or other health professional who carefully feels the breasts and underarm area for lumps or anything else unusual. Women may also perform a breast self-exam by checking their own breasts for lumps or changes in size or shape. The clinical breast exam and breast self-exam can help women become more familiar with the regular look and feel of their breasts and more readily identify changes
- **Mammography*:*** A low-dose x-ray exam that produces images of the breast called a mammogram. Mammography plays a central part in early detection of breast cancers because it can often show changes in the breast before a patient or physician can feel them. Research has shown that annual mammograms can lead to early detection of breast cancers when they are most curable and breast-conservation therapies are available. A mammogram may also find ductal carcinoma in situ (DCIS), abnormal cells in the lining of a breast duct that may develop into invasive cancer. Mammography is the only breast cancer screening tool known to reduce deaths from the disease.

Supplemental Breast Cancer Screening: Many studies have shown that ultrasound and magnetic resonance imaging (MRI) can help supplement mammography by detecting breast cancers that may not be visible with mammography. Neither MRI nor ultrasound is meant to replace mammography. Rather, they are used in conjunction with mammography in selected women. Women should consult with their referring doctor or radiologist to determine if MRI or ultrasound screening is appropriate for them.

- MRI of the breast may be used for women at high risk for breast cancer, typically because of a strong family history of the disease.
- Ultrasound of the breast may be used for women who are at high risk for breast cancer and unable to undergo an MRI examination or women who are pregnant and should not be exposed to x-rays used in mammography. Ultrasound of the breast is

also used to screen women who have dense breast tissue, meaning there are a lot of ducts, glands, fibrous tissue and less fat making it harder to find cancers with traditional mammography.

2.2.3 Biopsy of Breast cancer

A breast biopsy is a test that removes tissue or sometimes fluid from the suspicious area. The removed cells are examined under a microscope and further tested to check for the presence of breast cancer. A biopsy is the only diagnostic procedure that can definitely determine if the suspicious area is cancerous. There are three types of biopsies:

- **Fine-needle aspiration(FNA) biopsy:** In an FNA biopsy, a very thin, hollow needle attached to a syringe is used to withdraw (aspirate) a small amount of tissue from a suspicious area. The needle used for an FNA biopsy is thinner than the one used for blood tests.
- **Core-needle biopsy:** A core biopsy uses a larger needle to sample breast changes felt by the doctor or seen on an ultrasound, mammogram, or MRI. Ultrasound-guided breast biopsy uses ultrasound imaging to visualize a breast lump. Using real-time ultrasound images, a radiologist will advance a needle to the site of the abnormality and remove some tissue for evaluation under a microscope. The biopsy procedure is usually quick, but it may take a few days before the final tissue analysis (pathology report) is ready.
- **Surgical biopsy:** In rare cases, surgery is needed to remove all or part of the lump for testing. This is called a surgical or open biopsy. Most often, the surgeon removes the entire mass or abnormal area as well as a surrounding margin of normal breast tissue.

2.2.4 Breast Cancer Staging

Breast cancer stage is usually expressed as a number on a scale of 0 through IV with stage 0 describing non-invasive cancers that remain within their original location and stage IV describing invasive cancers that have spread outside the breast to other parts.

Stage 0 is used to describe non-invasive breast cancers, such as **DCIS (ductal carcinoma in situ).** In stage 0, there is no evidence of cancer cells or non-cancerous abnormal cells breaking out of the part of the breast in which they started, or getting through to or invading neighboring normal tissue.

Stage 1 describes invasive breast cancer in which is a small lump that is confined to the breast and should be eminently curable by removal of the breast or the part of the breast in which it is located.

Stage 2 breast cancer is when the primary cancer is less than 5 cm in diameter and has spread into adjacent lymph nodes that are still mobile but there is no evidence of spread beyond.

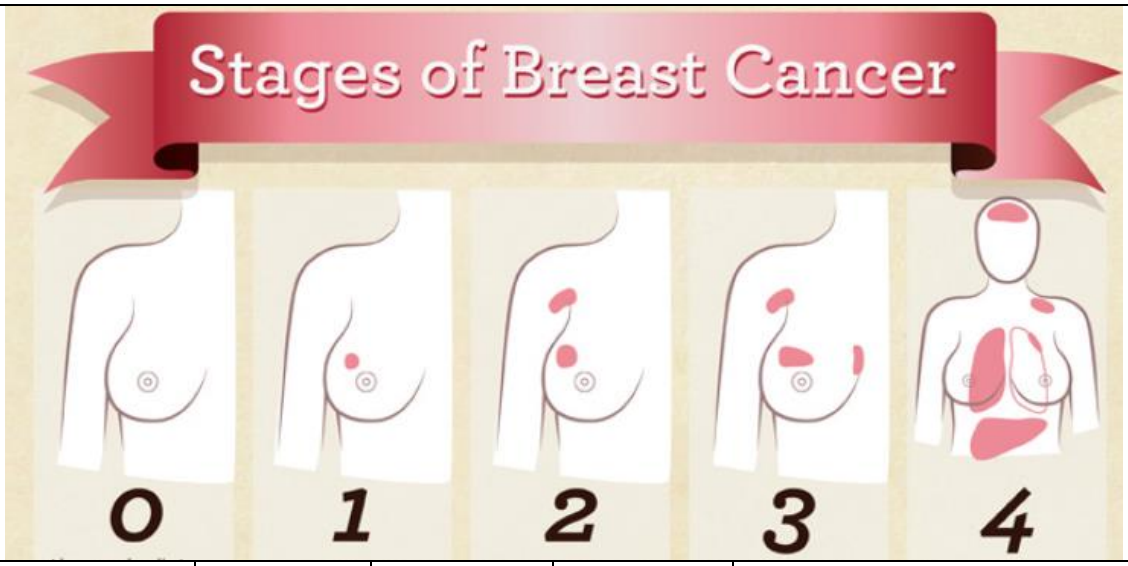
Stage 3 breast cancer is when there is a large cancer in the breast and/or involvement of overlying skin and/or involvement and adherence to underlying muscle and/or axillary lymph

nodes are adherent to surrounding tissues, but there is no evidence of spread into more distant organs or tissues.

Stage 4 breast cancer is when there is metastatic spread into more distant organs or tissues. These cancers are clearly not curable by treatment of the breast. The staging system of breast cancer is described in Table below.

TNM staging system: The most commonly used tool that doctors and researchers use to describe the stage is the TNM system. TNM stands for Tumor, Node, Metastasis [39] The figure 2.5 shows more details on each part of the TNM system for breast cancer: Doctors use the results from diagnostic tests and scans to answer these questions:

- **Tumor (T):** How large is the primary tumor? Where is it located?
- **Node (N):** Has the tumor spread to the lymph nodes? If so, where and how many?
- **Metastasis (M):** Has the cancer metastasized to other parts of the body? If so, where and how much?



100% Survival	98% Survival	88% Survival	52% Survival	16% Survival

Fig. 2.4 : Diagrams illustrating the four "stages" of breast cancer

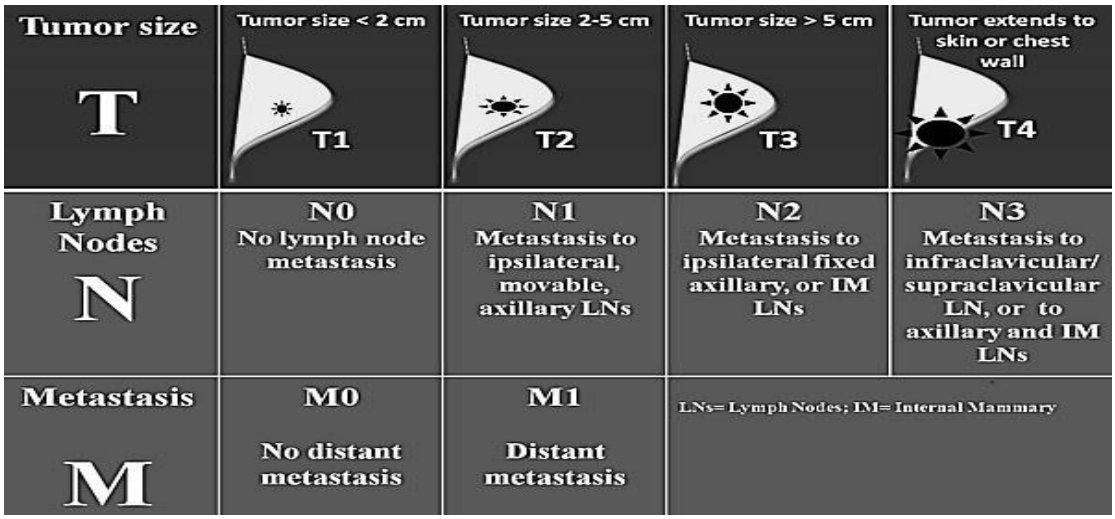


Fig. 2.5 :Diagrams illustration the TNM system of breast cancer

2.3 History of external beam radiotherapy in breast cancer

The journey of external beam radiation started in the last century with superficial X-rays and gradually moved on to high energy X-rays and photons including radionuclides like 60-Co (radioactive Cobalt) and 37-Cs (Cesium). The revolution in radiation oncology began post Second World War, with development of tele-cobalt followed by high energy linear accelerator in 1950s.However it was only in 1960–1970 when linear accelerators came more into practice for therapeutic medical indications. This period is thus popularly called megavoltage era [41]. Changes were notably apparent in technical aspects of radiation, starting from manual surface marking based planning to adoption of computer and software. The advent of computed tomography (CT) scan in the field of radiation oncology further added to the sophistication with its robust use in simulation, segmentation, dose and inhomogeneity correction which all together brought a significant change toward individual patient based planning compared with earlier 2D standard planning techniques. Modern linear accelerators are also equipped with different imaging facilities which made patient positioning and treatment delivery precise thus providing the opportunity of reducing target volumes. All of these essentially lead to a theoretical dogma of less radiation induced morbidity. Again, for breast cancer, most of which are detected very early with increased patient awareness, thus giving rise to the issue of long term survivorship, the use of new technology must be well planned and thought to avoid morbid survival. With the ray of hope in sight, thus, toxicity reduction, notably skin toxicity, cardiac toxicity and ischemic heart disease, pulmonary toxicity and risk of pulmonary fibrosis seem to be manageable and reducible. This again, has been said with caution, given the young age of these modern techniques and lack of data to prophesize the future.

2.4 Radiation Treatment Modalities

Breast cancer is commonly treated with radiotherapy. WBI has historically delivered 45 to 50 Gy in 25 to 28 fractions at 1.8 to 2 Gy per fraction. The close proximity of tumours to radiosensitive normal tissues means that, for many patients, successful cure is associated with sequelae of long-term radiation damage to these normal tissues. These include general tissue fibrosis and atrophy leading to stiffness of the tissues. Furthermore, several specific organ dysfunctions are observed. Before the time of IMRT and VMAT, conventional high energy photon treatment consisted of fixed source Co-60 (1.25 MV) treatment units which was developed during the 1950s, and linear accelerators in the 1960s. The main advantage of using megavoltage x-rays in simple beam deliveries such as relatively uniform and parallel opposed entrance beams is that the absorbed dose to the skin is reduced.

2.4.1 Conventional Radiotherapy

Conventional radiotherapy techniques have for many years used simple parallel opposed fields to treat breast cancer. Typically, treatment was planned using orthogonal plain radiographs using a simulator and field borders were based on standard anatomical bony landmarks. While these techniques provided adequate tumour coverage, there was little opportunity to spare adjacent normal tissues, leading to many of the side effects detailed above.

2.4.2 Three-Dimensional Conformal RT (3D-CRT)

Three-Dimensional Conformal Radiotherapy (3D-CRT) became available for the treatment of head and neck cancer in the 1990s. A radiotherapy technique where the dose volume is made to conform closely to the target through the use of 3D anatomical data acquired from CT or MRI imaging modalities. The aim is to apply the maximum dose to the target while sparing neighboring structures as much as possible with the aid of advanced computer software and hardware. 3D-CRT cannot always conform well to an arbitrary three dimensional shape unless a large number of beams are used or the target has a relatively simple shape. In many situations, the angles required to avoid or minimize dose to normal tissues are either difficult or impossible to implement clinically. Another limitation is if there are several critical structures surrounding the target, which is normally the case for many breast cancer sites. In this instance, there are few, if any, beam directions that can be placed strategically to satisfactorily avoid the sensitive normal tissues and appropriately irradiate the target. Finally, one of the greatest drawbacks to 3D-CRT is in situations where a concave tumor wraps around a sensitive structure, such as the spinal cord, heart, lung leaving no acceptable plan to be found without exceeding cord limits.

2.4.3 Intensity Modulated Radiation Therapy

Intensity Modulated Radiation Therapy (IMRT) is a modality built on the foundation of 3DCRT. It makes use of all the new planning tools that were introduced with 3DCRT when it was first employed in radiation oncology. A higher degree of conformity is achievable by

changing or modulating the intensity of each field. IMRT requires an inverse planning method which starts with splitting each field into very small segments commonly referred to as “beamlets.” Each of these beamlets is then given a different intensity in order to modulate the dose to produce a higher quality plan than what is achievable from 3DCRT alone. The intensity of each beamlet is determined by an automated optimization algorithm. The user chooses the number and geometry of the beams as well as target goals and constraints. The goals are desired objectives and the constraints act as limits. The automated optimization algorithm then attempts to create the best dose distribution given the information the user has provided. Although three dimensional conformal radiation therapy with its careful delineation of target and normal tissues and volumetric evaluation of dose, facilitated an increased target dose and or reduced normal tissue dose in certain sites, IMRT led to a considerably greater improvement in the therapeutic ratio. In IMRT treatment plan, the optimization parameters and structures are the primary variables used to control the dose distribution as opposed to the beam weights or shapes as in 3DCRT treatment planning. Specification of the optimization parameters and beam placement requires knowledge of how the details of the dose calculation algorithms and anatomical features of the patient such as the proximity of the normal tissues to the target affect the outcome of optimization. Hence, the combination of beams, optimization parameters, and structures needed to achieve the best treatment plan are patient specific [42].

2.5.4 Volumetric Modulated Arc Therapy (VMAT)

Arc- or rotational treatment therapy methods can be traced back to the midst of 20th century

and conformal arc therapy involving dynamic field shaping by MLCs were first described in the mid1960s, and developed further during the 1980s along with the three-dimensional conformal radiation therapy (3DCRT). Eventually, in 1995 Cedric Yu introduced the Intensity Modulated Arc Therapy, (IMAT) in which photon treatment was conducted in a rotational cone beam manner, where the modulation consists of different field shapes and varying dose weightings along the 360° rotation. With the prior knowledge of IMAT and by varying the dose rate of the linear accelerator, Karl Otto developed a single-arc IMAT algorithm in 2008 which he called Volumetric Modulated Arc Therapy (VMAT). The algorithm uses a progressive beam angle sampling to optimize a number of apertures, using direct aperture optimization, where the aperture shapes and their corresponding weights are initially optimized for a set number of coarsely placed gantrangles without taking the aperture connectivity in too much consideration. When the optimization algorithm has found an optimal solution, the previously left out gantry angles are inserted with their associated apertures being linearly interpolated from their neighbouring apertures to ensure a smooth transition of the MLC. Volumetric modulated arc therapy (VMAT) is a radiation technique in which the gantry moves continuously around the patient, while the radiation is constantly on. The gantry has varying speed, and the dose rate (MU per time) also varies **[43].** The shape of the treatment field changes dynamically during gantry rotation due to the varying velocities and positions of the MLC leaves, which can move back and forth along the y-axis. The result of a VMAT treatment is, as for IMRT, an intensity modulated dose distribution which conforms to the tumor volumes. A single rotation of up to 360◦ of the gantry is defined as an arc, and the number of arcs used for treatment is optional. Depending on the aperture models used for radiation delivery, different restrictions apply to gantry rotation, dose rate, and speed and position of MLC leaves.

2.5 Volume Definition

IMRT based treatment plans can have very sharp dose fall off away from the intended treatment volume. Therefore, positional accuracy of the treatment volume with respect to the beam geometry is of paramount importance. Several important concepts and standard terminology for this topic, as defined by the International Commission of Radiation Units and Measurements (ICRU) report no.50 and 62 are introduced in this section. These definitions, among other issues, specifically address the uncertainty in positioning the patient and the treatment volume. On an image, the Gross Tumour Volume (GTV) is the visible extent and location of the malignant growth and GTV may include the primary tumour site, metastatic lymph nodes and other metastases. However, due to limitations of imaging modalities, microscopic malignant cells cannot be adequately visualized, therefore, a clinical target volume (CTV) is defined by the physician to include both the GTV and the extent of microscopic diseases that surrounds it. The ideal goal of radiation therapy is to deliver a uniform, therapeutic dose of radiation to this volume spanned over a large number (typically 30) of daily fractions. Several factors introduce uncertainty in the radiation therapy process. They include 1) the daily set-up of patient, and 2) mechanical components of radiation therapy machine and 3) organ motions. These uncertainties make it difficult to align the CTV to the treatment beam in each fraction. To account for these uncertainties, a planning target volume (PTV) is defined to include the CTV and a margin to account for these errors, to ensure that the CTV will receive the prescribed radiation dose. Of the three factors that contribute to the geometric uncertainty, the mechanical tolerance of the radiation therapy treatment machine is the least contributing factor. Patient setup can be improved by employing some form of imaging which is, at least, capable of imaging bony anatomy of the patient in the treatment position. The most significant, and also the most difficult to circumvent, factor is the organ motion. It is important to distinguish between two types of organ motions: *inter*-fractional organ motion, which is the organ motion that occurs in the timescale of the days of treatment (such as a shrinking of the tumour as a response to treatment) an intra-fractional organ motion, which is the type of motion the occurs during treatment, such as organ motion that is associated with respiration. A PTV margins must account for all of these errors. A schematic diagram showing the 3 types of target volumes are shown in Figure 2.6.

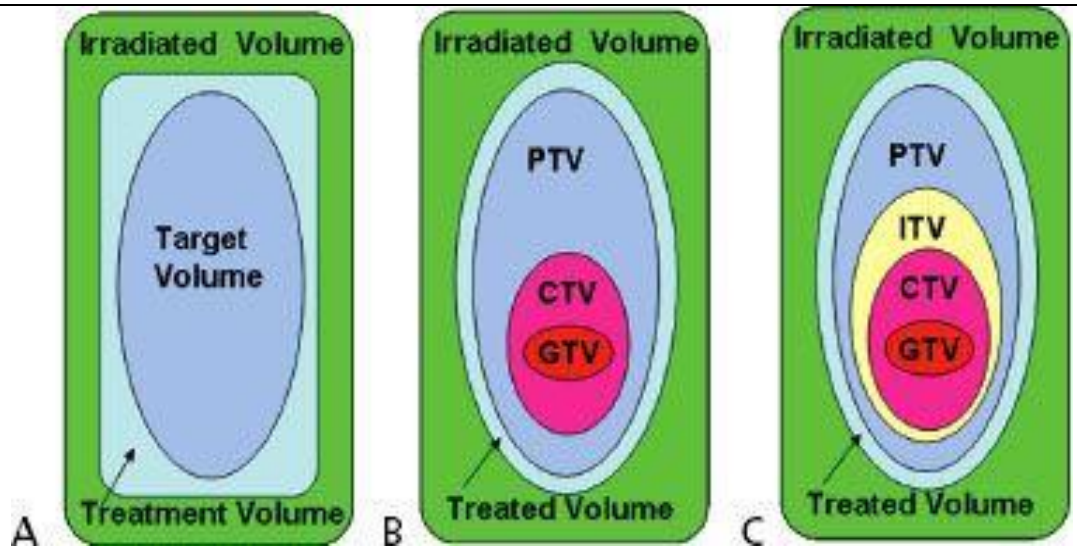


Fig. 2.6 Diagrams shows the ICRU 50 and 62

2.6 Whole Breast, Hypo-fractionated

Hypo-fractionation, or delivery of greater than standard 1.8 - 2 Gy fraction sizes per day, is a method of shortening overall treatment time in breast cancer. There are many potential benefits in delivering postoperative WBI in a shorter period of time. The advantages include greater convenience for patients, broad applicability to nearly all patients following lumpectomy, improved use of postoperative radiation for breast conservation, decreased treatment costs, and increased utilization of existing RT resources. Historically, standard fraction sizes of 1.8-2.0 Gy for radiotherapy were based primarily on studies examining squamous cell cancers from cervix and head and neck regions. The smaller fraction sizes exploited a biological differential in squamous cell cancer fractionation sensitivity versus normal tissue fractionation sensitivity. This allowed relative sparing of surrounding normal tissue from low dose per fraction. However, investigators from the United Kingdom hypothesized that the fractionation sensitivity for adenocarcinoma of the breast is close to that of the normal breast tissue and therefore with increasing fraction size a sufficiently large reduction of total dose could be implemented to keep late toxicity constant without losing tumor control.

2.6.1 Hypo-Fractionation and Concurrent Boost

There are 3 recent Phase I/II trials showing the safety and short-term efficacy of hypo-fractionated radiation therapy with a concurrent boost:

1) Freedman, et al. (2007) have reported a clinical study of hypofractionation using IMRT and an incorporated breast boost in early-stage breast cancer. Seventy-five patients were treated on study. The whole breast was treated to a dose of 2.25 Gy per day for 20 fractions for a total of 45 Gy. The incorporated boost gave simultaneously the tumor bed 2.8 Gy per fraction for 20 days for a total of 56 Gy. This use of hypofractionation of the whole breast volume, and simultaneously the boost volume, results in a 4-week overall treatment time.

2) Formenti, et al. (2007) have also reported a trial of IMRT, hypofractionation, and concomitant boost. A dose of 40.5 Gy was delivered in 15 fractions with a concomitant boost of 0.5 Gy per day for a total tumor bed dose of 48 Gy. The results in 91 patients treated were reported with a median follow-up of 12 months. The major acute toxicity was reversible grade 1-2 dermatitis in 67%. There were no treatment breaks. There were 2 acute grade 3 toxicities, 1 skin and 1 fatigue. There were no late grade 3 toxicities. Late fibrosis was reported grade 1 in 48%, grade 2 3%. Grade 1 pigmentation change was noted in 70%. Breast pain was grade 1 in 8% and grade 2 in 2%. Skin telangiectasias were grade 1 in 3% and grade 2 in 2%. There was 1 regional node recurrence.

3) Chadha, et al. (2009) have reported a trial of conventional whole breast irradiation with a concomitant boost over 3 weeks for early stage breast cancer. The whole breast dose was 2.7 Gy per fraction for 15 fractions to a dose of 40.5 Gy. The concomitant boost to the lumpectomy site was a total of 3 Gy per fraction for 15 fractions to a total dose of 45 Gy. The results of 105 patients were reported at a median follow-up of 24 months. There was no acute grade 3 or 4 toxicity. There were no reported late soft tissue toxicities. There was no significant negative effect reported on cosmesis 44].

2.7 Dose and Fractionation

WBI has historically delivered 45 to 50 Gy in 25 to 2 fractions at 1.8 to 2 Gy per fraction. However, 10-year data from multiple large randomized trials have demonstrated the non-inferiority of hypofractionated whole breast radiotherapy (42.5 Gy in 16 fractions (45) or 40.05 Gy in 15 fractions (46)). The Canadian trial included only node-negative patients and placed restrictions on breast size and treatment hot spots (46). The UK START trials had broader inclusion

criteria, including approximately 25% node-positive patients (47). Hypofractionated WBI should be strongly considered for appropriate patients, given the high-quality evidence supporting its safety and efficacy. This was highlighted in one of the ASTRO 2013 Choosing Wisely recommendations, stating not to initiate radiotherapy in women aged 50 years or older with early-stage invasive breast cancer without considering a hypofractionated treatment schedule (48). ASTRO has also released guidelines on dose and fractionation schedules in patients appropriate for hypofractionated radiotherapy (49). Prone positioning and advanced dose modulation strategies may help improve dose homogeneity and increase the number of patients appropriate for hypofractionation. More hypofractionated regimens are currently under investigation, such as 28.5 or 30 Gy in five once-weekly fractions of 5.7 or 6 Gy (50)

2.8 Hybrid IMRT Approaches

Treatment plans do not have to be constructed only for IMRT or only for VMAT, and many treatment centers combine IMRT and 3DCRT treatments in therapy for breast cancer [51]. In such cases, hybrid techniques typically concurrently combine 3DCRT fields (80% of the dose) and IMRT or VMAT fields (20% of the dose). Potential advantages of this type of treatments include a reduction in the volume of lung exposed to low doses. Hybrid-IMRT reduces treatment planning time, compared to full IMRT techniques, and minimizes the set

of optimization constraints. Hybrid-IMRT significantly reduces hot spot regions within the breast tissue and produces less skin reaction. The figure 1.7(A, B, C) shows more details on each part of the Hybrid-IMRT.

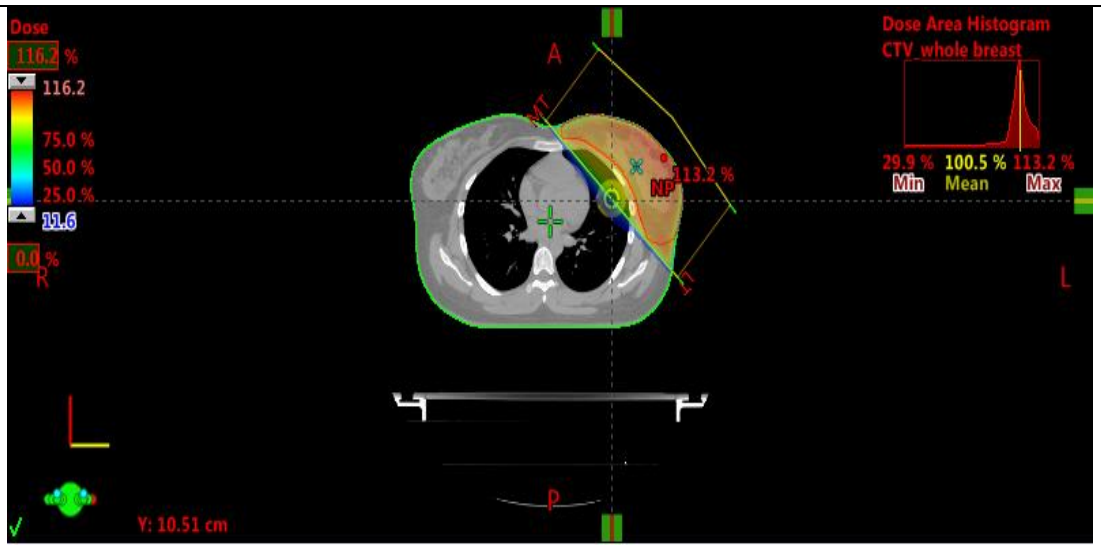


Fig. 2.7(A) Diagrams shows the 3DCRT plan of Hybrid-IMRT plan

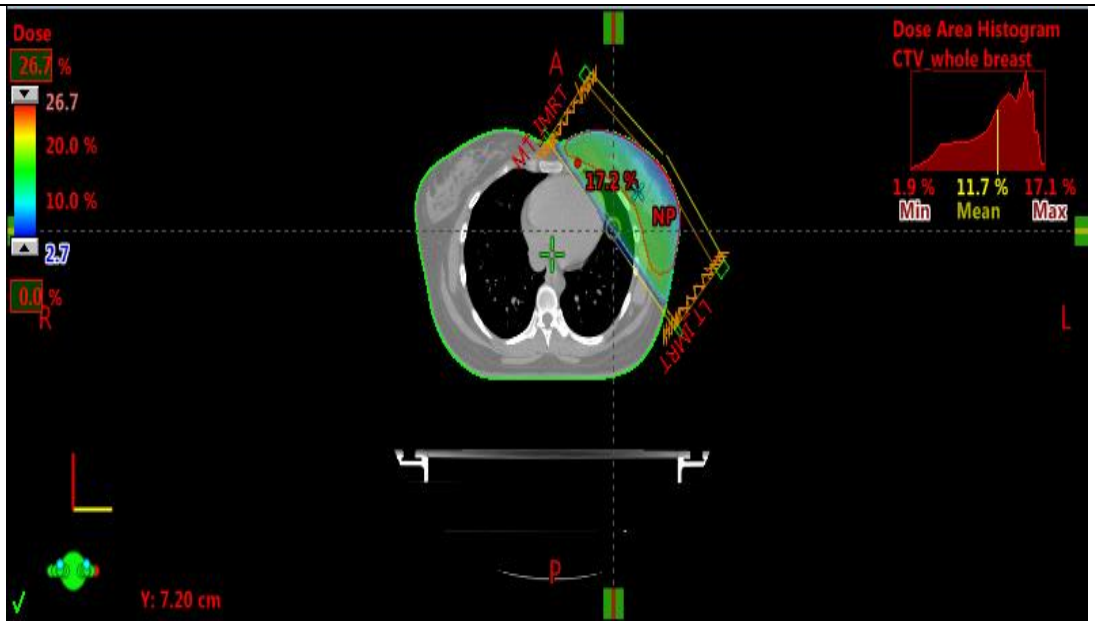


Fig. 2.7(B)Diagrams shows the IMRT plan of H-IMRT plan

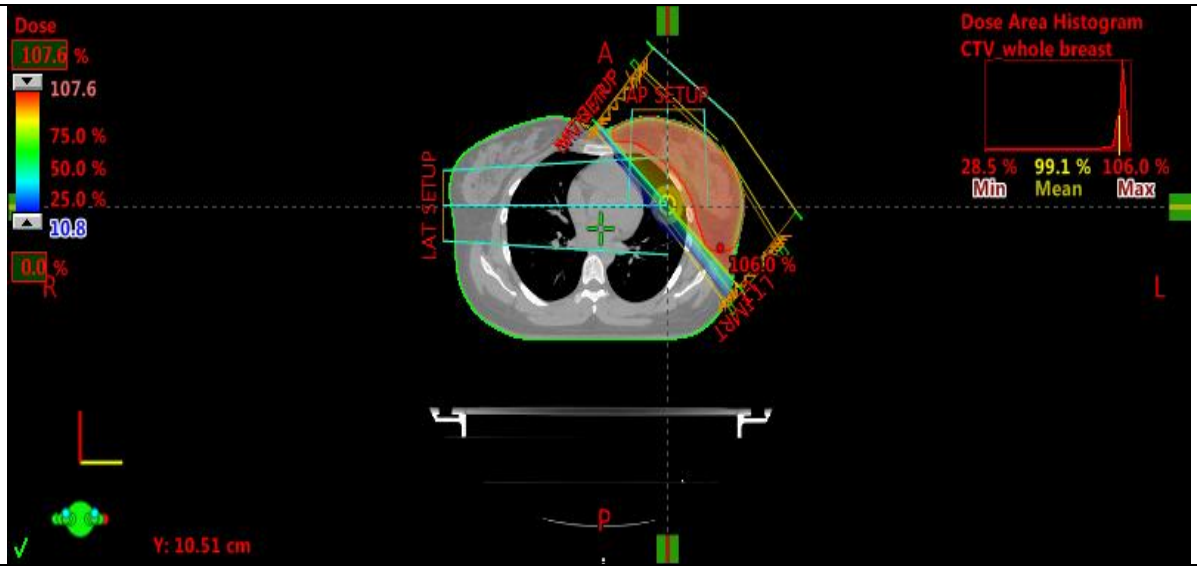


Fig.2.7(C) shows the combination of 3DCRT and IMRT which is called H-IMRT plan.

Chapter-3: Materials and Methods

3.1 Materials

The following materials are used for this study.

1) Patient Selection

2) CT-Simulator and Immobilization

3) Varian Eclipse Treatment Planning System (V13.7).

4) Protocols (EMAMI and QUANTEC, RTOG 1005, RTOG-1304 and ICRU-83)

A short description of these materials are given below:

3.1.1 Patient Selection

Ten patients were enrolled in this retrospective study under an Institutional Review Board-approved protocol. These patients had left sided, early-stage invasive mammary carcinoma (pT1N0M0), and underwent breast-conserving surgery followed by radiotherapy at the hospital between January 2018 and April 2018. Patients between 34 and 46 years old, with adequate function of the liver, kidney, heart, and hematopoietic system were considered eligible for the study. Patients with positive axillary or supraclavicular lymph nodes and distant metastasis were excluded from the study. All patients were immobilized in the supine position with the arm abducted (90° or greater) on the disease side. A computed tomography (CT) scan with 5 mm slice thickness was acquired for each patient, with coverage from the mandible to 4–6 cm below the inframammary fold to cover the entire lung volume. Imaging data were then transferred to the Eclipse treatment planning system (Varian Medical Systems, USA) for delineation of targets and critical structures.

3.1.2 CT Simulation and Immobilization.

Computed tomography (CT) simulator combines functionality of a conventional simulator with features and image processing and display tools of a three-dimensional radiation therapy treatment planning (3D RTTP) system. Radiotherapy treatment simulation devices have been an integral component of treatment planning and delivery process for over 30 years. Image-guided three-dimensional radiation therapy is increasingly becoming a standard of care in radiation oncology community. Figure 3.1 shows the Siemens CT simulator which has the Flat Tabletop of “CT Overlay” provides a secure, flat surface for CT Simulation applications, consistent with the treatment couch for reproducible patient position

Carbon fiber construction with foam core provides a durable, lightweight device with outstanding imaging properties.



Figure 3.1 shows the Siemens CT simulator

AIO cover image breast board is designed to achieve a comfortable and reproducible supine position of the patient (figure 1.9). This position is primarily used for the treatment of malignancies in the thorax and the abdomen and breast cancer. The set consists of wedges, with different angles to increase patient comfort, and arm rests to allow for a comfortable but reproducible positioning of the arms above the head and out of the treatment fields. Hand grips and head supports will further help in increasing the precision and in finding the correct patient position. 5°, 10° and 15° angled wedges lift the upper body of the patients and thereby facilitate breathing. The coating of the cushions prevents the patient from sliding down. The armrests (20° and 30° angle) have a flat top to support the total upper arm. This increases the patient's comfort.



Fig. 3.2 shows the breast board for the breast cancer.

3.1.3 Eclipse Treatment Planning System

Eclipse treatment planning system (V13.7) was designed and developed by Varian Medical Systems. The Varian Eclipse™ Treatment Planning System (Eclipse TPS) provides software tools for planning the treatment of malignant or benign diseases with radiation. Eclipse TPS is a computer-based software device used by trained medical professionals to design and simulate radiation therapy treatments. The Eclipse Treatment Planning System (Eclipse TPS) is used to plan radiotherapy treatments for patients with malignant or benign diseases. Eclipse TPS is used to plan external beam irradiation with photon, electron and proton beams, as well as for internal irradiation (brachytherapy) treatments. In addition, the Eclipse Proton Eye algorithm is specifically indicated for planning proton treatment of neoplasms of the eye. In a Varian environment, Eclipse integrates the ARIA oncology information system making workflow smooth, integrated and seamless. As Varian takes radiation treatment to the next level, Eclipse is designed to advance along with it.

3.1.4 Protocols

The following protocols have been used for this study.

- EMAMI and QUANTEC.
- RTOG 1005 and 1304
- ICRU 83

3.1.4.1 EMAMI and QUANTEC

Dr. EMAMI and Quantitative Analysis of Normal Tissue Effects in the Clinic (QUANTEC) Protocol are used in the Radiation Oncology. The QUANTEC effort was made possible, in part, by generous financial support from the American Society for Radiation Oncology (ASTRO) and the American Association of Physicists in Medicine (AAPM). The results were published in March 2010 in a special issue of the *International Journal of Radiation Oncology·Biology·Physics* (the Red Journal). Table 3.1. Shows Normal Tissue Tolerance for Standard Fractionation:

Table 3.1: - Tolerance level of Normal Tissue.

Organ	Endpoint	Rate (%)	Dose-volume parameter	D_{mean}(Gy)
Heart	Pericarditis Cardiac Mortality	<15% <1%	V30<46% V25<10%	<26Gy
Lung	Pneumonitis	20%	V20<=30%	20Gy

3.1.4.2 RTOG-1005

The Radiation Therapy Oncology Group (**RTOG-1005**)1005 is “A Phase iii trial of accelerated whole breast irradiation with hyrofractionation plus concurrent boost versus

standard whole breast irradiation plus sequential boost for early stage breast cancer". This thesis has been carried out to follow the RTOG-1005 (Fig. 2.0- A, B)

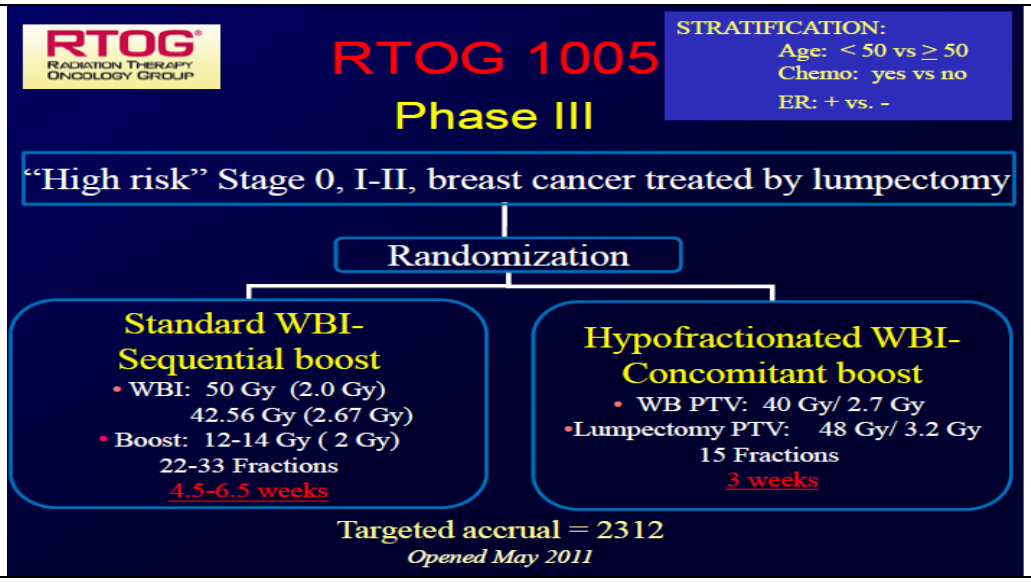


Figure 3.3-A explains the RTOG-1005 with Hypo-fractionated WBI.

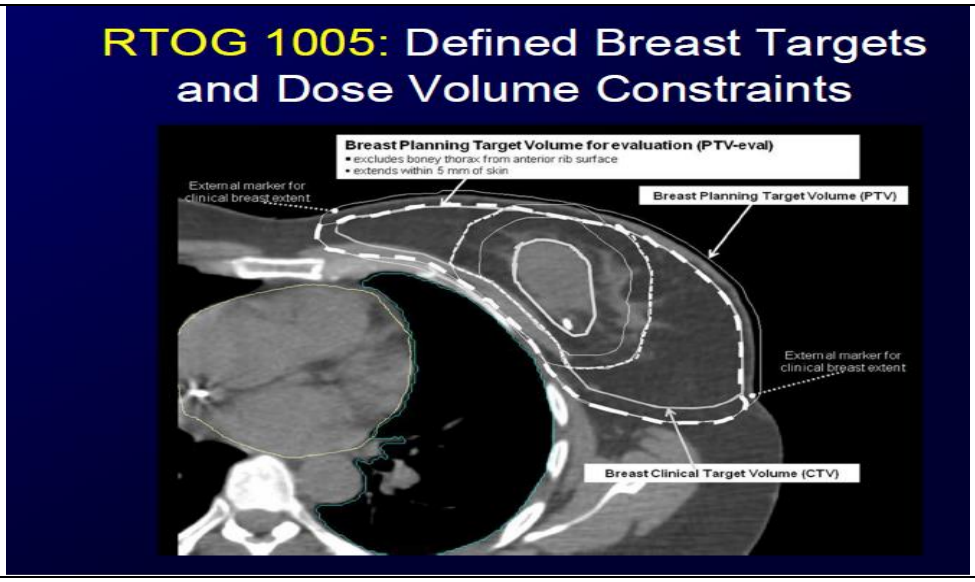


Fig.3.4 Illustration of Post Lumpectomy Target Delineation

3.1.4.3 RTOG- 1304

NSABP B-51/RTOG 1304 is a randomized phase III clinical trial evaluating regional nodal (RN) radiotherapy (RT) in patients with positive axillary nodes before neo-adjuvant chemotherapy (NC) who convert to negative axillary nodes (ypN0) after NC. Normal Tissue Constraints and Goal from NSABP B-51/RTOG 1304 illustrated in table 3.2. RTOG-1304 (Fig.3.5 A, B) illustrated the delineation of Target volume and OAR of Breast cancer.

Table 3.2: - Constraints and Goal on NSABP B-51/RTOG 1304

Organ	Ideal	Acceptable
Heart Mean dose	≤ 4Gy	≤ 5Gy
Heart, Volumetric Constraints(Left)	V25Gy ≤ 5%; V15Gy ≤30%	V30Gy ≤ 5%; V15Gy ≤35%
Heart, Volumetric Constraints(Right)	V25Gy ≤ 0; V15 ≤ 10%	V30Gy ≤ 05; V15 ≤ 15%
Ipsilateral Lung	V20Gy ≤ 30% V10Gy ≤ 50% V5Gy ≤ 65%	V20Gy ≤ 35% V10Gy ≤ 60% V5Gy ≤ 70%
Contralateral Lung	V5Gy ≤ 10%	V5Gy ≤ 15%
Contralateral Breast	V3Gy ≤ 5%	V4.1Gy ≤ 5%
Breast PTV Eval	95%	90%
Chest wall	≥95% receives 47.5Gy	≥90% receives 45Gy
Axilla	≥95% receives 47.5Gy	≥90% receives 45Gy
SCV	≥95% receives 47.5Gy	≥90% receives 45Gy
IMN Nodes	≥90% receives 45Gy	≥90% receives 45Gy

Vx refers to the volume of the target volume receiving x% of the dose (i.e., V90 ≥ 90.0% means that greater than 90.0% of the volume was covered by 90.0% of the prescription dose). VxGy refers to the volume of the target volume receiving × Gy (i.e., V5Gy ≤ 15.0% means that less than 15.0% of the volume was covered by 5 Gy).

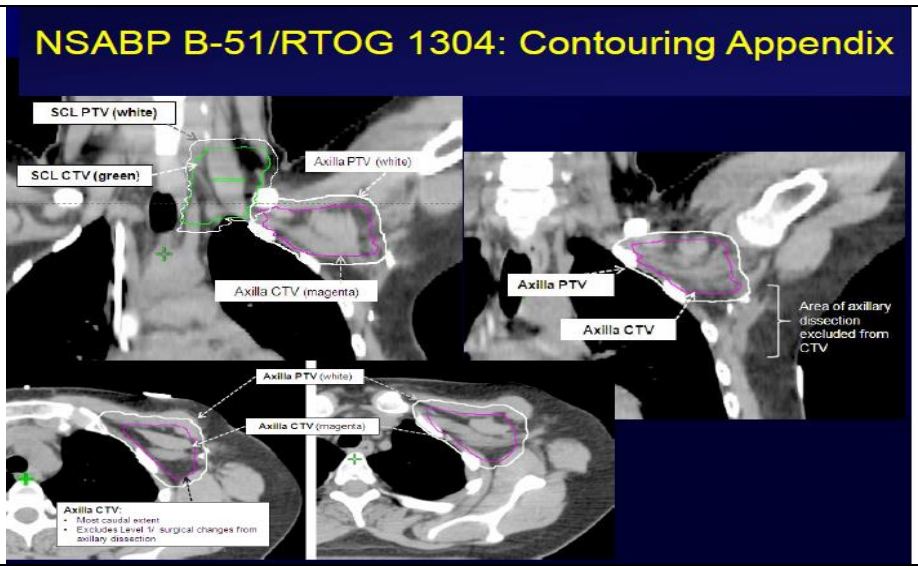


Fig. 3.5-B: Supra-clavicle delineation for nodal spread case

3.1.4.4 ICRU 83 protocol

The ICRU 83 Report was released in 2010 specifically addressing IMRT and introducing some different concepts and plan evaluation parameters such volumetric PTV prescribing. In this research the treatment plans were reviewed using ICRU-83 recommendations. The Homogeneity Index (HI) and Conformity index (CI) were also calculated according to ICRU-83 recommendations.

3.1.4.4.1 Dose Homogeneity and Conformity Index

Dose homogeneity and dose conformity are independent specifications of the quality of the absorbed dose distribution. Dose homogeneity characterizes the uniformity of the absorbed dose distribution within the target volume. Dose conformity characterizes the degree to which the high-dose region conforms to the target volume, usually the PTV. Figure below illustrates examples of dose homogeneity and dose conformity [52].

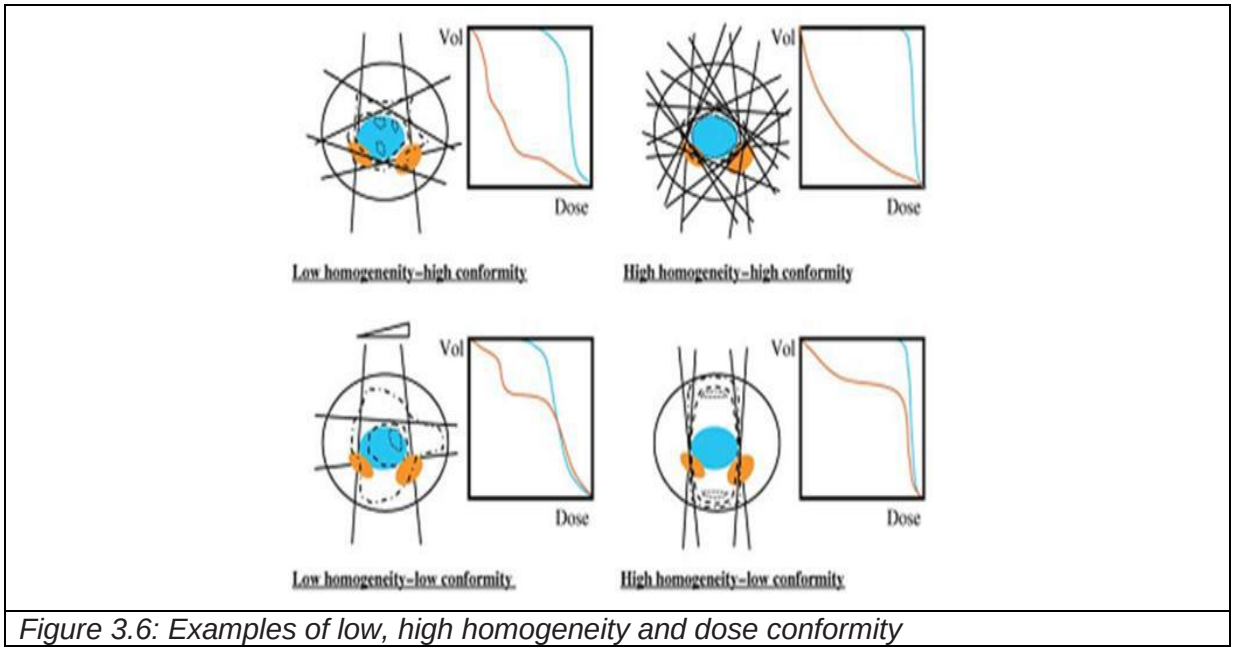


Figure 3.6: Examples of low, high homogeneity and dose conformity

The Conformity index (CI) according to the ICRU-83 is

CI ref = (Volume of PTV covered by the reference dose) /Volume of PTV. CI= 1.0 for ideal case. According to the ICRU-83 the Homogeneity index (HI) is defined as

HI = (D2%-D98%) /D50% Where, D 2%, D 98%, D 50% is dose received by 2%, 98%, 50% volume. HI = 0 (Zero) is ideal val.

3.2 Methods

The practical work for this study was carried out by the following parameter which is given below:

1) Delineation of target volumes and organs at risk
2) Dose and Fractionation
3) Treatment planning

3.2.1 Delineation of Target Volumes and Organs at Risk

All patients were immobilized in the supine position on a AIO cover image breast board which increased patient comfort, and arm rests to allow for a comfortable but reproducible positioning of the arms above the head and out of the treatment fields. A CT scan with a slice thickness of 5 mm was acquired from each patient with coverage from the mandible to 4 to 6 cm below the inframammary fold to cover the entire lung volume. After CT scan, Digital Imaging and Communication in Medicine (DICOM) images were transferred to Eclipse treatment planning system (V13.7). The breast targets with other volume was delineated by the radiation oncologist with the following considerations. The clinical target volume (CTV) for the whole left breast included the supraclavicular head (Fig. 3.2-D) as the superior margin, 2 cm below the inframammary fold as the inferior margin, anterior axillary line as the lateral margin, and midsternal line as the inner margin. The CTV for the tumor bed was defined as the lumpectomy cavity (seroma) found on the CT scan image or the fibrous tissue under the surgical scar if no seroma could be found on the CT scan image. The normal tissue and OAR including healthy tissue, the lungs, the heart, the spinal cord, and the contralateral breast were contoured for dose calculation. The target delineation was performed based Radiation Therapy Oncology Group (RTOG) guidelines (Fig.3.2. A, B, C, D and Fig.3.3) The margin from CTV to PTV was 5 mm, except for superficial areas where PTV was never closer than 5 mm to the skin. The body was delineated on the CT scans and Boolean operations were used to construct a modified body volume that operations were used to construct a modified volume that excluded breast PTV.

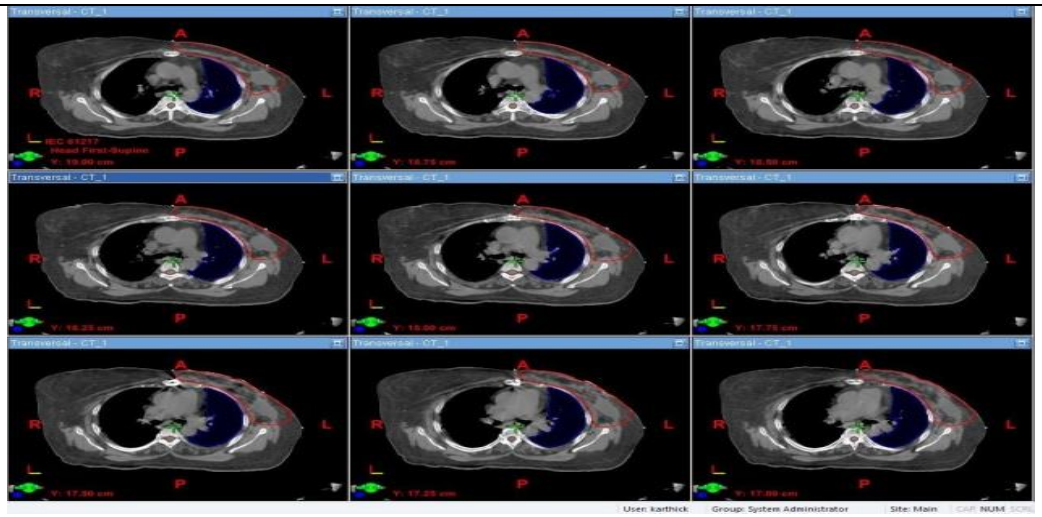


Fig.: 3.7 A. Illustration of Breast Post Mastectomy Target Delineation

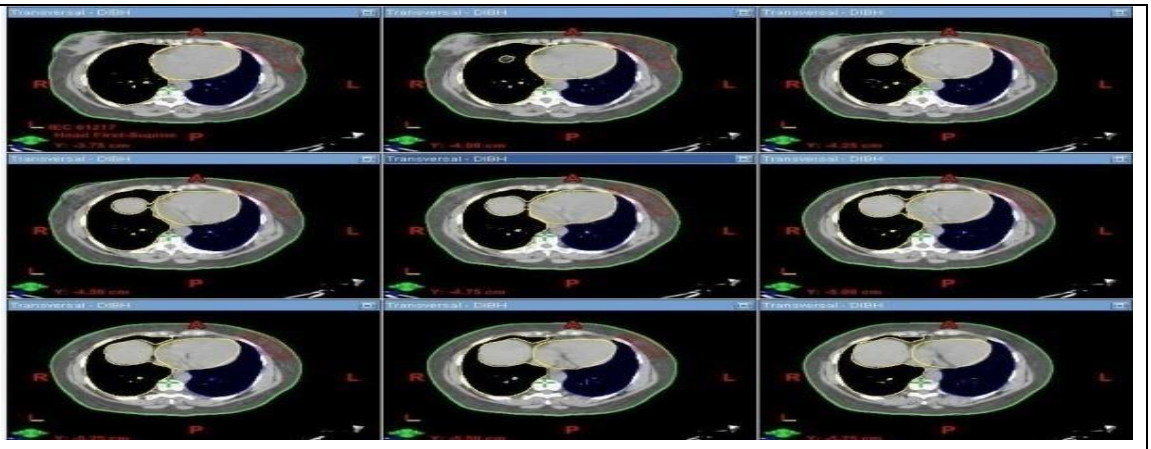


Fig.: 3.7B. Illustration of Post Lumpectomy Target Delineation

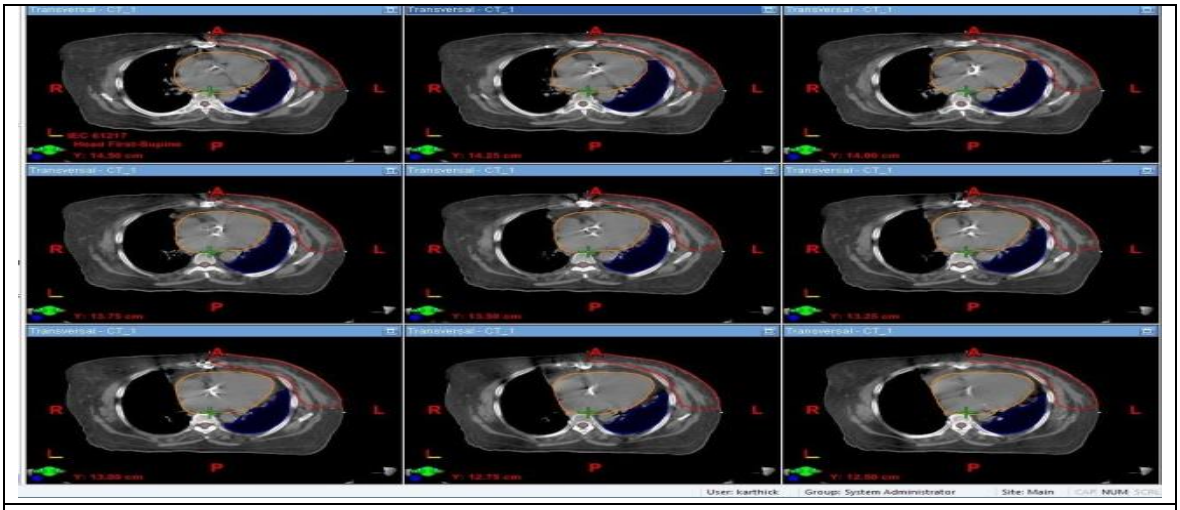


Fig. 3.8 C Target, Left Lung and heart delineation

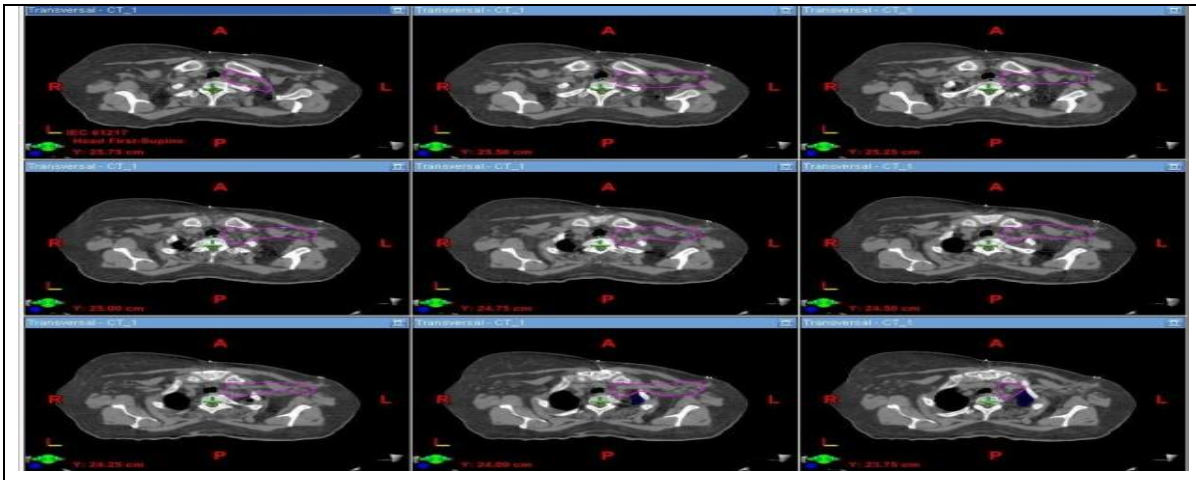


Fig. 3.9 D Supra-clavicle delineation for nodal spread case

Breast/ Chestwall Contour: Anatomical Boundaries

	Cranial	*Caudal*	*Anterior*	*Posterior*	*Lateral*	*Medial*
Breast	Clinical Reference + Second rib insertion[a]	Clinical reference + loss of CT apparent breast	Skin	Excludes pectoralis muscles, chestwall muscles, ribs	Clinical Reference + mid axillary line typically, excludes lattismus (Lat.) dorsi m. [b]	Sternal-rib junction [c]
Breast-Chestwall[2\]	Same	Same	Same	Includes pectoralis mucles, chestwall muscles, ribs	Same	Same
Chestwall[3]	Caudal border of the clavicle head	Clincial reference+ loss of CT apparent contralateral breast	Skin	Rib-pleural interface. (Includes pectoralis mucles, chestwall muscles, ribs)	Clinical Reference/ mid axillary line typically, excludes lattismus dorsi m [a]	Sternal-rib junction [b]

Fig.3.10 Illustration of Breast/Chest wall Contour Anatomy

3.2.2 Prescribed dose and Fractionation

Standard WBRT consists of 45–50 Gy in 25–28 fractions of 1.8–2 Gy. Long-term data from multiple large randomized trials also demonstrate the non-inferiority of hypo-fractionated WBRT (HF-WBRT) consisting of 40.05–42.56 Gy in 15–16 fractions of 2.66–2.67 Gy for early stage breast cancer with equal or lesser acute and long-term toxicity. **The prescription dose to the whole breast was 40.05 Gy in 15 fractions of 2.67 Gy in this study according to the RTOG- 1005.** The Boost dose of 10 Gy at 2 Gy per fraction was delivered

to the tumor bed after delivery of 40.05 Gy using 6 MV Photon beams to the entire breast. Boosting to a total tumor bed dose greater than 50.05 Gy should be considered in patients with positive margins**.**

3.2.3 Hybrid-IMRT Treatment Planning Techniques

For each patient, CT scans were contoured by one radiation oncologist and were carried out by one medical physicist to ensure plan uniformity. Hybrid IMRT plan was executed, and the afflicted breast was exposed to 40.5 Gy in 15 fractions of 2.67 Gy to cover the PTV. All the treatment plans were generated using an Eclipse treatment planning system(V13.7). The treatment plans were planned to deliver with 6-MV photon beams on a 2300 C/D linear accelerator that is mounted with a 120-leaf of Millennium multi-leaf collimator (MLC) (maximum leaf speed of 2.5 cm/s). While executing the plan on each patient, the same isocenter and tangential beams were applied. The hybrid technique consisted of a two-step process in which one step is 3DCRT and second step is IMRT.

In the first planning step (3DCRT), two opposing 6 or 10 MV tangential fields [Fig.3.11] at one isocenter were added conforming the PTV-$_{Breast}$. A margin of 3cm anteriorly was added. This ensures entire breast coverage in spite of breathing. The beam angles and beam weighting (usually minimal) were chosen to optimize coverage of the PTV, while minimizing exposure to the ipsilateral lung, heart and contralateral breast Gantry angles ranged from 310° to 350° for the medial fields and from 110° to 125° for the lateral fields for patients treated on the left side. The fields extended 2 cm anteriorly of the chest to provide coverage of the “flash” region. All other plans were normalized to achieve isodose coverage of the breast tissue at least as good as the tangent plan. A portion of the total breast dose in 3DCRT, that is, 80%, was calculated to a reference point as per ICRU 50 and 62 guidelines, and weightings were used to provide an even distribution of dose across the whole breast

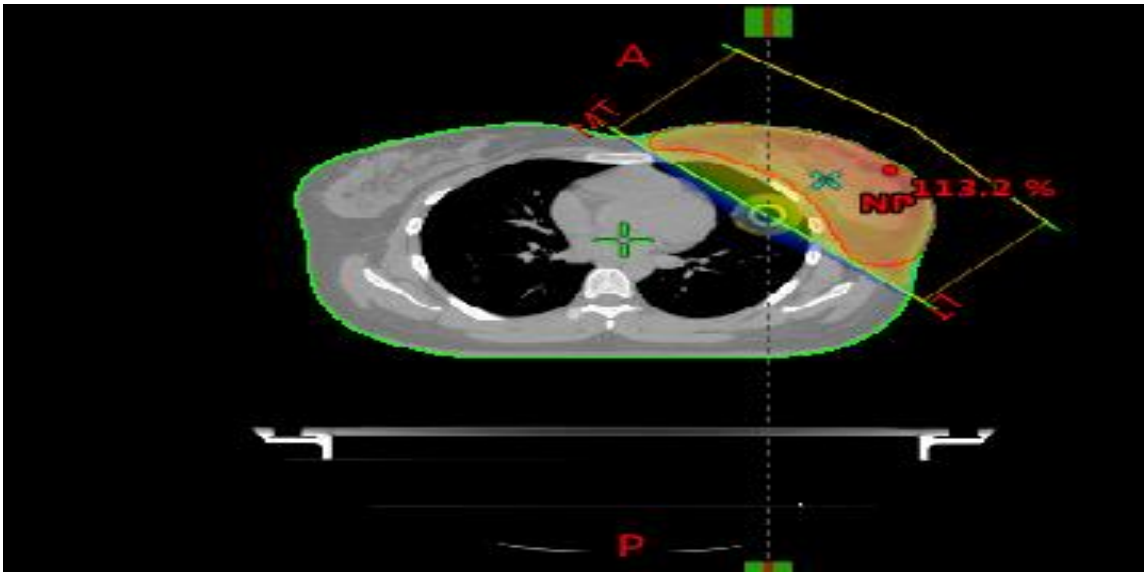


Fig.3.11 Illustration of 3DCRT of Hybrid IMRT

As the second step, two IMRT fields [Fig. 3.12] were added at the same isocenter; two tangential fields at the same gantry angle as the original two tangents and two fields angled to best access the boost cavity. The two non-tangential fields were dedicated to treat reflecting a conventional tangent plan. A portion of the total breast dose in IMRT, that is, 20%, was calculated to a reference point as per ICRU 83 guidelines, and weightings were used to provide an even distribution of dose across the whole breast. No additional fields or beam modifiers were used in this plan. A second IMRT plan was created at the same isocentre as the conventional plan and comprised of two fields; two tangential fields at the same gantry angle as the original two tangents and two fields angled to best access the boost cavity. The two non-tangential fields were dedicated to treat the boost cavity by locking the jaws around the PTVboost during the optimisation process. The IMRT plan was optimised using a standard optimisation template employing sliding window dynamic IMRT with Eclipse optimization software to achieve dose constraints and utilise avoidance volumes to reduce undesirable excess dose. A basic level of planning experience was simulated by limiting optimisation calculations and use of avoidance structures to two. The plan was dosed with the remaining, that is, 20%, of the total breast dose and 10 Gy additionally for the boost cavity.

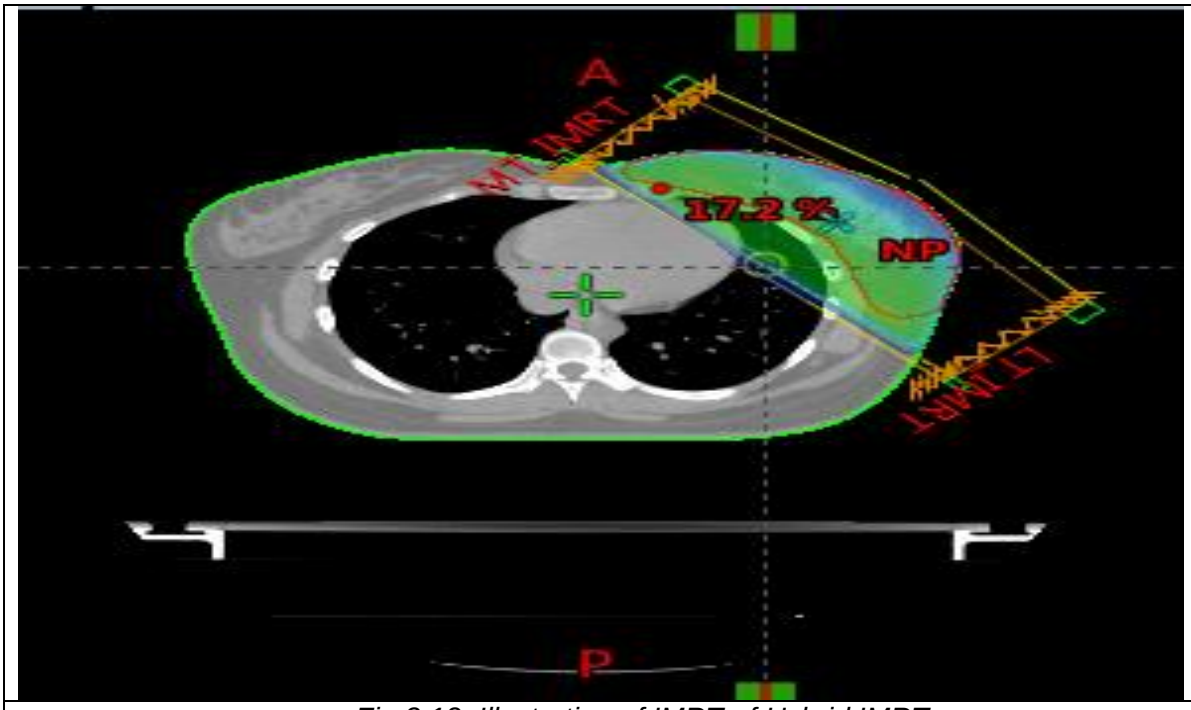


Fig.3.12. Illustration of IMRT of Hybrid IMRT

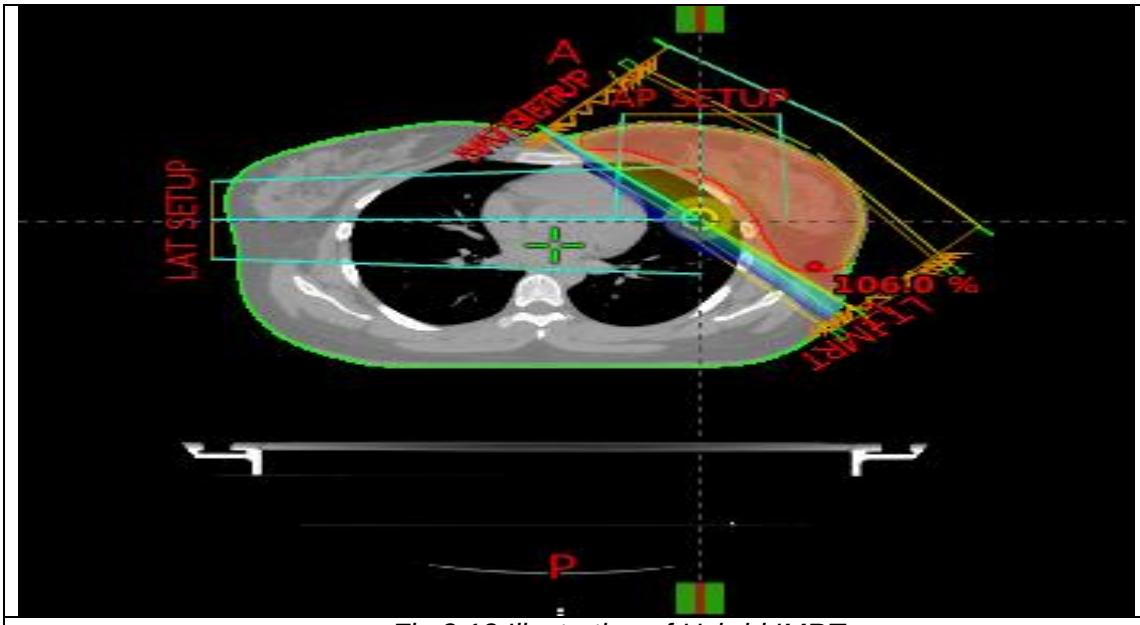


Fig.3.13 Illustration of Hybrid IMRT

For the hybrid-IMRT treatment plan, 2 tangential IMRT fields and 3DCRT fields were integrated to a single treatment plan [Fig. 3.13]. The standard medial and lateral 3D-CRT beams with control points were created by adding a 2.5 cm expansion margin on the surface of the chest wall skin, and a 0.5 cm margin was added in the other directions. Two IMRT beams had the same gantry angles as that of the tangential beams. The beam angles and field weights were chosen to optimize coverage of the PTV, while minimizing exposure to the left lung, heart and right breast. This study h followed QUANTEC and RTOG protocols.

Chapter-4: Results

4.1 Dose Analysis of Target Volume

Dose distribution color washes from the treatment modality of H-IMRT for two representative patients in this study. The patients were chosen for their dosimetric parameters most closely matched the mean value of the eight patient's data set. Figure 4.1 and 4.2 shows axial and frontal dose distributions with H-IMRT using by colour wash. H-IMRT achieved 95% coverage of the CTV_ whole breast. We found that H-IMRT deliver fewer doses to the OAR and improve better dose coverage across the whole breast for all patients in this study. The mean volumes, averaged from the 8 patients, for CTV_ whole breast, heart, left lung and right lung were 1140 ±104, 526.4±132, 1050 ± 210, and 1205±205.7 (cm3), respectively.

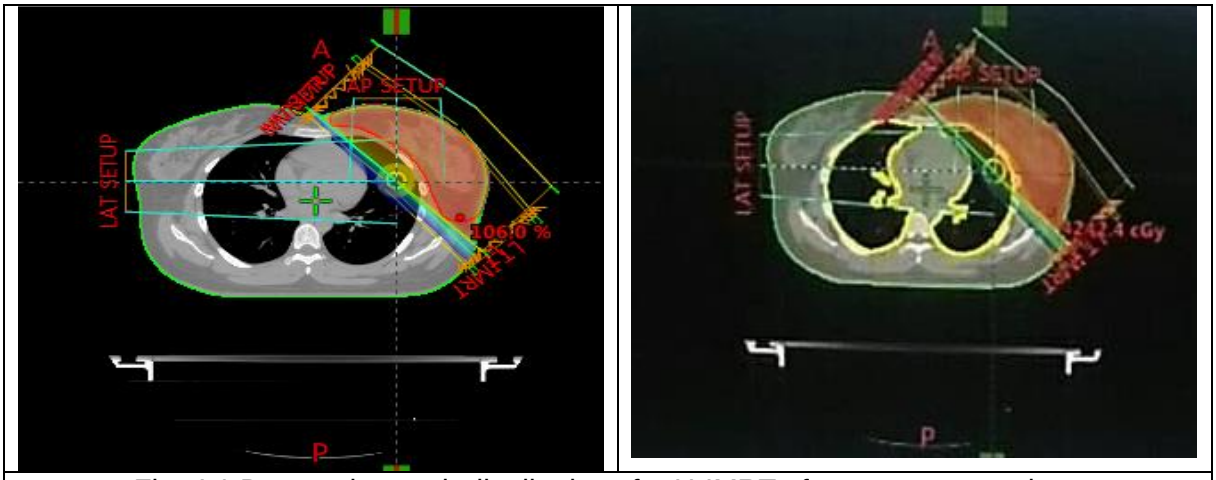


Fig. 4.1 Dose-color wash distributions for H-IMRT of two treatment plans

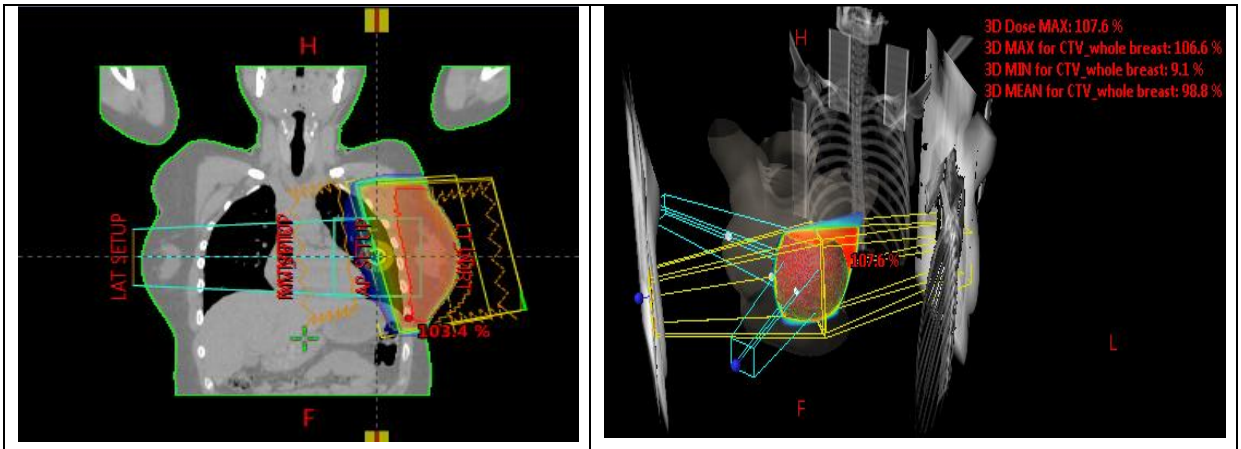


Fig. 4.2 Dose-color wash for H-IMRT of frontal view

4.2 Dose Analysis in OAR (Heart)

As shown in table 4.1 CT-simulation scans demonstrated significantly irradiated heart volume for H-IMRT. H-IMRT also resulted in a significant reduction in the heart V5, V20 and V30 and mean heart dose. The mean dose to the entire heart was smaller for the H-IMRT plans according to protocol RTOG. Table 4.1 also shows that; heart V5 is 3.55% which value is significantly lower in the hybrid-IMRT plans compared to protocol RTOG.

Table.4.1: Dose characteristics of the Heart of H-IMRT including the mean dose for left-sided patients.

Structure	Parameter	H-IMRT	Acceptable value	Ideal value
Heart mean dose	D_{mean}(Gy)	(1.64 ± 0.55)	≤ 5Gy	≤ 4Gy
Heart Volumetric Constraint (Left)	V5(%)	3.55 ± 2.65	V5 ≤ 35%;	V5 ≤ 30%;
	V20(%)	2.5 ± 0.55	V20Gy ≤30%;	V20≤ 25%;
	V30(%)	1.05 ± 0.65	V25Gy ≤ 5%;	V30Gy ≤ 5%;

4.3 Dose and Volume Analysis in OAR (L-Lung and R-Lung)

The D mean and V Gy (the volume of the organ receiving at least a given dose (Gy)), for the left and right lungs, esophagus, and spinal cord in the H-IMRT are shown in Table 4.2. The hybrid–IMRT plans produced the best results with low dose area to OAR and better parameters of the contralateral lung and ipsilateral lung. The hybrid–IMRT plans contributed less percentage of ipsilateral lung receiving a radiation dose more than 20Gy (V20Gy) compared to protocol.

Table.4.2: Characteristics of dose directed to lung, Contralateral lung in H-IMRT for left-sided patients.

Structure	Parameter	H-IMR mean	Acceptable value	Ideal value
Ipsilateral Lung	Dmean (Gy)	7.5 ± 2.24	13	7
	V5Gy (%)	28% ± 4.5%	V5Gy ≤ 70%	V5Gy ≤ 65%
	V10Gy (%)	20.24% ± 5.6%	V10Gy ≤ 60%	V10Gy ≤ 50%
	V20Gy (%)	14.77% ± 6.9%	V20Gy ≤ 35%	V20Gy ≤ 30%
	V30 Gy (%)	11.98% ± 2.8%	V30Gy ≤ 66%	V30Gy ≤ 60%
Contralateral Lung	Dmean (Gy)	0.097		
	V5(%)	0.0%±0.0%	V5Gy ≤ 15%	V5Gy ≤ 10%

4.4 Dose and Volume Analysis in OAR (R-Breast, Esophagus)

Table 4.3 presents the irradiation doses and volumes in the contralateral breast, esophagus and spinal cord of H-IMRT plans. The hybrid–IMRT plans produced the best results with low dose area to OAR and better parameters of the contralateral breast and esophagus and spinal cord.

Table.4.3: Dose characteristics of the right breast, esophagus and spinal cord of H-IMRT including the mean dose for left-sided patients.

Structure	Parameter	H-IMR mean	Acceptable value	Ideal value
Contralateral Breast	Dmean(Gy)	0.5 ± 0.3		
	V5(%)	0.1% ± 0.2%	V4.1Gy ≤ 5%	V3Gy ≤ 5%
Esophagus	Dmean(Gy)	0.447±0.1		
Spinal cord	Dmean(Gy)	0.182 ±0.0		

4.5 Dose volume histogram for a patient

Dose volume histogram (DVH) serves as an important tool for evaluation of treatment of treatment plans for organs with variable densities. Dose volume histogram (DVH) for CTV_ whole breast and organs at risk (OAR) structures for all eight patient's data sets are same which are displayed in figure 4.2 by one patient. A DVH used clinically includes all structures of interest in the radiotherapy plan. The horizontal axis represents the dose (cGy) received, while the vertical axis lists the percentage of the total volume that receives the specified dose. In figure 4.3 shows the mean dose volume histogram (DVH) for CTV_ whole breast, lung, esophagus and heart which are better coverage according to protocol.

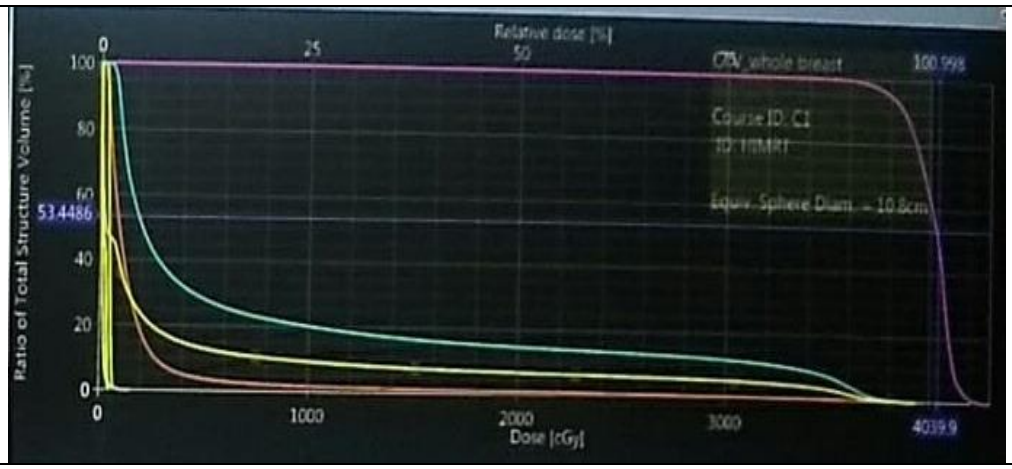


Fig. 4.3: Dose volume histogram (DVH) for one patient

Chapter 5: Discussions

5.1 Discussions

Many studies have shown benefits of H-IMRT in left breast patients, especially in minimizing the cardiac complications. This study was undertaken to appraise the benefit of H-IMRT compared to RTOG protocol for locally advanced left sided chest wall breast patients. Early studies indicated that H-IMRT could reduce the volume of the heart within tangential fields. Discussion of the results for H-IMRT compared with literature of 3DCRT, IMRT and VMAT has been given below:

Rahbin et al [20] have shown that, the homogeneity of dose to planning target volume (PTV) and the dose delivered to heart and contralateral breast were compared among the techniques in all the 20 patients. All the three radiotherapy techniques achieved comparable radiation dose delivery to PTV-95% of the prescribed dose covering > 95% of the breast PTV. The mean volume of PTV receiving 105% (V 105) of the prescribed dose was 1.7% (range 0-6.8%) for IP-IMRT, 1.9% for FP-IMRT, and 3.7% for 3D-CRT. The low-dose volumes (V 5Gy) in the heart and lungs were larger in IP-IMRT than in the other techniques. The value of the mean dose to the ipsilateral lung was higher for IP-IMRT than the values for with FiF-FP-IMRT and 3D-CRT. It provided dosimetric advantages, significantly reducing the size of the hot spot and minimally improving the coverage of the target volume. In addition, it was felt that FiF-FP-IMRT required less planning time and easy field placements. The results are slightly different compared to this study. In this current study, the data shows that tangential beam H-IMRT for left breast cancer patients offers the potential to significantly reduce dose-volume parameters of the mean heart dose, heart V5Gy, V20, V30 and left lung parameters (V5, V10, V20 and V30) contralateral breast respectively compared to RTOG and **QUANTEC** protocols with comparable target dose coverage.

In the article, Gamal Elhusseiny, MD et al [21] demonstrated that IMRT becomes more suitable for treating complex target volumes as it is able to generate more complicated dose distributions and dose gradients with narrower margins than those allowed using traditional methods. As a consequence, IMRT can provide a clinical benefit in terms of increased tumor control through escalated dose and reduced normal tissue complications through OARs sparing. **In this present study** firstly proved that hybrid IMRT plans requiring less MUs can achieve dosimetric equivalence to IMRT. These hybrid IMRTs are also more flexible in terms of positioning repeatability. In the case of hybrid IMRT, the dosimetric equivalence is achieved by combining the same amount of IMRT beams to the standard tangential conformal beams. In this manner, hybrid IMRT plans not only reduce hot spots outside the target volume but also improve dose homogeneity within the target volume.

In the article, Sherif Elzawawy et al. [22] compared the dosimetry for the left breast cancer radiotherapy using three different radiotherapy techniques, tangential wedged fields (TW), segmented field (FiF) and inverse planning IMRT (IP-IMRT). Regarding organs at risk, left lung and heart have higher values of V5, V10, and V20 for IP-IMRT compared to TW and FiF; the differences are statistically significant, lower coronary artery regionV30 vules for IPIMRT compared to TW and FiF but no difference in the Dmean between IPIMRT and FIF. FiF and TW decrease the contralateral breast dose significantly compared to IP IMRT.

In this present study, Noticeably, for hybrid IMRT, the PTV volume received 95 or 90% prescription dose: this was significantly higher than for the article of Sherif Elzawawy. The hot spots outside the target were also evaluated. The hybrid IMRT plans were comparable

with the IP-IMRT plans, but they were better than the 3D-CRT plans in restricting the volumes of tissue outside the target from receiving ≥100 and ≥110%, prescribed dose. H-IMRT for left breast cancer patients produces the potential to significantly reduce dose-volume parameters of the mean heart dose, heart V5Gy, V20, V30 and left lung parameters (V5, V10, V20 and V30) contralateral breast respectively.

In the article, Suresh Moorthy et al. [23] demonstrated that SIB-3DCRT treatment plans were created for 36 consecutive patients. Dose was prescribed as 45 Gy in 25 fractions to the planning target volume (PTV)-1 and 60 Gy in 25 fractions to PTV-2. Treatment plans were normalized to 95% of PTV volume receiving 95% of the prescription dose. The conformity index (CI), homogeneity index (HI), lung dose, heart dose, left anterior descending artery(LAD) dose, and low dose volume and integral dose of normal healthy tissue were recorded and analyzed. However, there was no significant difference in the HI ($p = 0.45$). On average, the V20Gy of ipsilateral lung was 37.9 % for 3DCRT and 22.4 % ($p < 0.01$) for IMRT, whereas the V20Gy of total lung (ipsilateral + contralateral) was 21.8% for 3DCRT and 12.14 ($p < 0.01$) for IMRT. Similarly, average V40Gy of heart was 7.5 % for 3DCRT and 2.13 % ($p = 0.01$) for IMRT. The LAD maximum dose to left side breast patients, on average, was 39.5 Gy for 3DCRT and 29.17Gy ($p = 0.03$) for IMRT.

In this current study, the main goal of the hybrid IMRT plan for left breast cancer is to protect the heart, both the lungs, and the contralateral breast during postoperative radiotherapy to the left breast. Hybrid IMRT average mean heart dose was (1.64 ± 0.55) Gy, V5(%) was 3.55 ± 2.65, V20 (%) was 2.5 ± 0.55 & V30 (%) was 1.05 ± 0.65 respectively. Hybrid IMRT average mean left lung dose was (7.5Gy ± 2.24) Gy. V5 (%) was 28% ± 4.5%, V10 was 20.24% ± 5.6%, V20 (%) was 14.77% ± 6.9% and V30(%) was 11.98% ± 2.8% respectively.

In the article, Azza Helal et al. [24] have demonstrated that various radiation techniques include Three Dimensional Conformal Radiotherapy (3DCRT), breath hold and Intensity Modulated Radiotherapy (IMRT) are used to irradiate the breast and to limit the dose to the heart and lungs. 3DCRT and IMRT plans were generated for 50Gy using opposed tangential medial and lateral photon beams. Five segments were generated for each IMRT field. Heart V5Gy, V20Gy, V30Gy and V40Gy and ipsilateral lung V20Gy, mean dose, PTV D95%, PTV Dmax and dose homogeneity index were compared using Wilcoxon Signed-Ranks test. IMRT plans achieved significant reduction of 12%, 15%, 21% & 53% in heart V5Gy, V20Gy, V30Gy and V40Gy and of 18% & 11% in ipsilateral lung mean dose and V20Gy compared with 3DCRT. **In this current study,** Hybrid IMRT average mean heart dose was (1.64 ± 0.55) Gy, V5(%) was 3.55 ± 2.65, V20 (%) was 2.5 ± 0.55 & V30 (%) was 1.05 ± 0.65 respectively. Hybrid IMRT average mean left lung dose was (7.5Gy ± 2.24) Gy. V5 (%) was 28% ± 4.5%, V10 was 20.24% ± 5.6%, V20 (%) was 14.77% ± 6.9% and V30(%) was 11.98% ± 2.8% respectively. The most important advantage of H-IMRT is treatment delivery time better than IMRT.

In the article, Charles S Mayo et al. [28] have evaluated a hybrid intensity modulated radiation therapy (IMRT) technique as a class solution for treatment of the intact breast. The following five plan techniques were compared for 10 breast patients using dose–volume histogram analysis: conventional wedged-field tangents (Tangents), forward-planned field-within- field tangents (FIF), IMRT-only tangents (IMRT tangents), conventional open plus

IMRT tangents (4-field hybrid), and conventional open plus IMRT tangents with 2 anterior oblique IMRT beams (6-field hybrid). The 4-field hybrid and FIF achieved dose distributions better than Tangents and IMRT tangents. The volume of tissue outside the planning target volume receiving >110% of prescribed dose was largest for IMRT tangents (average 158 cc) and least for 6-field hybrid (average 1 cc); the FIF and 4-field hybrid were comparable (average 15 cc). Heart volume >30 Gy averaged 13 cc for all techniques, except Tangents, for which it was 32 cc. Average total lung volume >20 Gy was 7% for all. Contralateral breast doses were <3% for all. Planning time for hybrid techniques was significantly less than for conventional FIF technique. The 4-field hybrid technique is a viable class solution. The 6-field hybrid technique creates the most conformal dose distribution at the expense of more normal tissue receiving low dose.

The similar results from this study show that H-IMRT significantly reduce the mean heart dose, the mean lung dose, lung parameters and Contralateral breast. This study also evaluated the delivery time with H-IMRT technique.

In the article, Ahmad Amoush et. al. [32] have compared the single-isocenter, four field hybrid IMRT with the two-isocenter techniques to treat the whole breast and supraclavicular fields and to investigate the intra-fraction motions in both techniques in the superior direction. The percentage of the total lung volume receiving > 20 Gy (V20Gy) and the heart receiving > 30 Gy (V30Gy) were 13.6% vs. 14.3% (p = 0.03) and 1.25% vs. 1.2% (p = 0.62), respectively. Shifting the treatment isocenter by 1 mm and 2 mm superiorly showed that the average maximum dose to 1 cc of the breast volume was 55.5 ± 1.8 Gy and 58.6 ± 4.3 Gy in the two-isocentric technique vs. 56.4 ± 2.1 Gy and 59.1 ± 5.1 Gy in the single-isocentric technique (p = 0.46, 0.87), respectively. The single-isocenter technique using four-field hybrid IMRT approach resulted in comparable plan quality as the two-isocentric technique. The single-isocenter technique is more sensitive to intra-fraction motion in the superior direction compared to the two-isocentric technique. The advantages of the single-isocenter include elimination of isocentric errors due to couch and collimator rotations and reduction in treatment time. This study supports consideration of a single-isocenter four-field hybrid IMRT technique for patients undergoing breast and supraclavicular nodal irradiation.

The similar results from this study show that H-IMRT significantly reduce the mean heart dose, the mean lung dose, lung parameters and Contralateral breast. This study also evaluated the delivery time with H-IMRT technique.

In the article, OUYANG Shuyu et. Al. [33] evaluated the potential dosimetric benefits and optimal indications of intensity modulated radiation therapy (IMRT) and hybrid intensity modulated radiation therapy (Hybrid IMRT) for the left side breast cancer patients after breast-conservation therapy. The study was compared with the Hybrid IMRT and IMRT: V13 of the ipsilateral lung decreased from 27.66% to 20.7%, V5 of the contralateral lung decreased from 8.01% to 2.25%, V10 and V20 of the heart decreased from 35.23% and 16.77% to 19.22% and 10.6% respectively, V5 and V10 of the contralateral breast decreased from 35% and 10.39% to 20.38% and 5.7% respectively, all with significant difference. V30 and V40 of the ipsilateral lung and V40 of the heart increased by 1.28%, 1.48%, and 2.48%, with significant difference. Hybrid IMRT is a better choice for patients whose treatment position is inaccurate or cannot be repeated well.

The similar results from this study show that H-IMRT significantly reduce the mean heart dose, the mean lung dose, lung parameters and Contralateral breast but lung parameters increased than the article, OUYANG Shuyu.

In the article, Osamu Takahashi**1** et. al. [35] have demonstrated to quantify the target coverage, homogeneity, and robustness of the dose distributions against geometrical uncertainties associated with four whole breast radiotherapy techniques. With the original plan, the wedge technique yielded a high volume receiving ≥107% of the prescription dose (V107; 7.5%±4.2%), whereas the other three techniques yielded excellent target coverage and homogeneity. A 5 mm anterior displacement caused a large and significant increase in the V107 (+5.2%±4.1%, p<0.01) with the FIF plan, but not with the hybrid IMRT (+0.4%±1.2%, p=0.11) or full IMRT (+0.7%±1.8%, p=0.10) plan. A 5-mm posterior displacement caused a large decrease in the V95 with the hybrid IMRT (-2.5%± 3.7%, p<0.01) and full IMRT (-4.3%±5.1%, p<0.01) plans, but not with the FIF plan (+0.1%±0.7%, p=0.74). The decrease in V95 was significantly smaller with the hybrid IMRT plan than with the full IMRT plan (p<0.01). The FIF, hybrid IMRT, and full IMRT plans offered excellent target coverage and homogeneity. Hybrid IMRT provided better robustness against geometrical uncertainties than full IMRT, whereas FIF provided comparable robustness to that of hybrid IMRT.

The similar results from this study show that H-IMRT significantly reduce the mean heart dose, the mean lung dose, lung parameters and Contralateral breast. This study also evaluated the delivery time with H-IMRT technique. However, hybrid IMRT offered excellent target coverage and homogeneity comparable to that of full IMRT.

In the article, Jia-Fu Lin et al. [38] compared the dosimetric performance of 3 different treatment techniques: hybrid volumetric-modulated arc therapy (hybrid-VMAT), pure-VMAT, and fixed-field intensity-modulated radiation therapy (F-IMRT) for whole-breast irradiation of left-sided early breast cancer. plan quality. Total monitor units (MUs) and delivery time were also recorded and evaluated. The hybrid-VMAT plan generated the best results in dose coverage of the target and the dose uniformity inside the target ($p < 0.0001$ for conformal index [CI]; $p = 0.0002$ for homogeneity index [HI] of planning target volume [PTV] (50.4 Gy) and $p < 0.0001$ for HI of PTV (62 Gy)). Volumes of ipsilateral lung irradiated to doses of 20 Gy (V (20 Gy)) and 5 Gy (V (5 Gy)) by the hybrid-VMAT plan were significantly less than those of the F-IMRT and the pure-VMAT plans. The volume of ipsilateral lung irradiated to a dose of 5 Gy was significantly less using the hybrid-VMAT plan than that using the F-IMRT or the pure-VMAT plan. The total mean MUs for the hybrid-VMAT plan were significantly less than those for the F-IMRT or the pure-VMAT plan. The results of the current study were supported by Jia-Fu Lin et al. [38] but the current study was not match with the results of Jia-Fu Lin's article.

Chapter 6: Conclusion

6.1 Conclusion

The objective of this study was to investigate the dosimetric characteristics of H-IMRT technique to the left-side breast radiotherapy in the early stage of the breast Cancer. In the current study, the data shows that tangential beam H-IMRT for left breast cancer patients offers the potential to significantly reduce dose-volume parameters of the ipsilateral lung, heart and contralateral breast compared to tangential beam 3-DCRT and IMRT of articles with comparable target dose coverage. In this study, we first proved that hybrid IMRT plans requiring less MUs can achieve dosimetric equivalence to full-IMRT. This hybrid IMRT is also more flexible in terms of positioning repeatability. In the case of hybrid IMRT, the dosimetric parameters equivalence are achieved to VMAT beams to the standard tangential conformal beams. In this manner, hybrid IMRT plans not only reduce hot spots outside the target volume but also improve dose homogeneity within the target volume. However, H-IMRT technique was superior in terms of low dose volume of normal tissue, integral dose, and treatment time. In context of limiting mean and low dose to lung and heart, H-IMRT is found to achieve good OAR sparing with acceptable PTV coverage for free breathing, left breast irradiation. So from the overall consideration we suggest to choose H-IMRT which was suitable for the protection of heart in the left-sided breast cancer radiotherapy.

Chapter 7: References

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